

ARTICLE

Double descent and statistical learning theory

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Abstract

The analysis of learning action, machine learning and related practices in the field theoretically has been made by utilizing Computational Learning Theory (CoLT) Valiant 1984 and Statistical Learning Theory (SLT) Vapnik 1999. These two theories, while overlapped, provides a general framework in analysing and justifying learning actions and learning model constructions, aiding in the formation of modern practices. One of the famous insight using such framework is the *bias-variance tradeoff* Geman, Bienenstock, and Doursat 1992, which states that model complexity and generality inversely affect each other, thus guarantee the need for a safe bracket between them. However, recent literatures, Belkin et al. 2019a has indicated the fallout of such dilemma by the new phenomenon called *double descent*, where there exists an interpolation threshold that renders the current statistical justification for bias-variance inconsequential to a given region. Various ‘anomaly’ has also been detected similarly, with varying degree of sophistication and potential. In this paper, we would like to review through progress made in this particular field of research, and the phenomena potential explanations up until now.

Keywords: Statistical learning theory, double descent, deep learning, computational complexity theory, complexity theory, grokking

The modern machine learning landscape is dominated and guaranteed by the foundation of theoretical machine learning, specifically computational learning theory (CoLT) and statistical learning theory (SLT). The theory of machine learning and its modern practice has been developed and researched, of a substantial portion by empirical and heuristic approach, either by advancing new practices, architectures or by try-and-test modification. From its early onset of a regression estimator $\theta(x, y)$ on the linear regression problem, machine learning has developed substantially. Certain model concept with great successes includes regression-classification model, Bayesian modelling, generative learning model, support vector machine, Gaussian processes, and more. Of all such, the more formal and complex model architecture created, is the concept of a *neural network*. With increasingly sophisticated architecture, heuristic approach become popular, the fast-paced advancement of the field comes with new method, new results, new observations, and its far-reaching application which led to even bigger and larger scale deployment, there has been questions about the formation and status of a theoretical ground, a rigorous matter on the side of **theoretical machine learning**.

While rigorous and well-formulated in a sense, classical and theoretical machine learning was dwarfed by the modern advancement of machine learning as a whole, leading to several anecdotal problems regarding the interpretation of phenomena, the re-evaluation of the theory to fit the more updated analysis, and explanation to more sophisticatedly designed system. This and many more, plus as present, many of such advancements and improvements are heuristic, and the general theory and conceptual understanding remain limited, led to the choice to often opt for analogies and empirical workaround. Ultimately, this resulted in multiple ‘failures’ in

explaining, interpreting, and predicting behaviours of new phenomena appeared in the modern landscape of machine learning. While there exists novelty of analysis, and different theoretical schemes to analyse such, particular problems remain in the form of irregularity and conflict of interpretations between frameworks.

Our main focus is to review on the topic, analyse existing analysis and observations, and understanding of the umbrella term classical learning theory, for the recent phenomena observed named ‘double descent’. Statistical learning theory (SLT) and computational learning theory (CLT) Vapnik 1999; Mohri, Rostamizadeh, and Talwalkar 2012; Shalev-Shwartz and Ben-David 2014; Hajek and Raginsky 2021; Bousquet et al. 2020 has been prominent in constructing a well-rounded formal theory surrounding learning problems, models, and machine learners analysis. Valuable insights have been dissected from treatment of statistical theory and mathematical modelling on models, including the *bias-variance tradeoff* Geman, Bienenstock, and Doursat 1992; P. M. Domingos 2000b, which serves as a bound for efficient learning and model configuration (or complexity). However, recently there has been observations of *double descent* Belkin et al. 2019a; Schaeffer et al. 2023; Nakkiran et al. 2019; Lafon and Thomas 2024 which refute the famous tradeoff assumption, and hence brings question to the establishment of the theory, as well as several assumptions and insight in the framework. Further events and phenomena observed also includes grokking and triple, to n -descent Davies, Langosco, and Krueger 2023; Ascoli, Sagun, and Biroli 2020. Furthermore, many problems of defining and formalizing notions used in designing and implementing machine learning models are inconclusive, such as, for example, *model complexity* and others.

1. Outline

The main contributions of the paper lie in the heart of the problem of *machine learning theoretical*, and specifically *double descent*. As such, we focus on the following:

1. Theoretical analysis of the machine learning framework, and reinterpretation of the empirical-generalization framework. We explicitly redetermine perspectives and formulation, about the problem of generalization learning, and model selection principles. Additionally, we also formalize the algorithmic objective of the objective system.
2. Observations and insight reviews and critical analysis. We analyse current knowledge, different analysis of the phenomena, and provide critiques, critical analysis on limitation, constraints, different assumptions and fallacy observed in those results, and verification or extension of experiments. This means analysing existing papers specific of details, and intrinsic of certain functional class.
3. Providing our own insight to the phenomena of double descent, and hence the need for new analysis or different theory scheme to be utilized in solving the problem.

While this is non-conclusive of the problem of double descent, we want to emphasize the importance of such issue to be further than simply the dilemma between model complexity and risk error. It hints at the mismatch of knowledge, fuzzy nature of certain concepts, failure modes being apparent that indicates the intricacy of the sensitivity of the theory when encountering general structures, and the immaturity in which the theoretical landscape contains many noises.

2. Related works

On itself, the phenomena *double descent* has been investigated somewhat comprehensively, firstly introduced by Belkin et al. 2019a, which gives it distinctive saddle escape illustration. Further analysis was made by several literatures, particularly attributed the existence of double descent to the concept of model complexity and inductive bias, and potential explanations of such. Nakkiran et al. 2019 expanded the phenomena into deep neural network models. Their conclusion is reached by considering the perturbation of a learning procedure $\mathcal{T}$ on the effective model complexity $\text{EMC}_{\mathcal{D},\epsilon}(\mathcal{T})$ defined in the paper, separating the eventual phenomena into regions of observations, thereby in one way or another, predicting the tendency of double descent. This is also the first, arguably, "concise" definition of double descent, even though its nature is an empirical definition, including the notion of model complexity. Recently, there is also the Neural Tangent Kernel (Jacot, Gabriel, and Hongler 2018) which can be used to explain double descent, however further researches are still required. Preliminary works are done by Lafon and Thomas 2024, Schaeffer et al. 2023, and Liu and Flanigan 2023 on the role of optimization in double descent. Davies, Langosco, and Krueger 2023 attempted to unifying grokking with double descent, theorized a possibility of similarity during the generalization phase transition of the inference period, and Olmin and Lindsten 2024 attempted to explain epoch-wise double

descent within the model of two-layer linear network. However, it is mentioned that it can also be expanded into deep nonlinear networks.

Further literatures on double descent varies, but fairly new, ever since their discovery articles in Belkin et al. 2019a and Nakkiran et al. 2019. Since then, we have Shi et al. 2024; Schaeffer et al. 2023; Quéty and Tartaglione 2023a; Lafon and Thomas 2024; Neal 2019; Davies, Langosco, and Krueger 2023; Quéty and Tartaglione 2023b; Liu and Flanigan 2023; Mei and Montanari 2020, 2019b; Adlam and Pennington 2020; Transtrum et al. 2025; Liu, Liao, and Suykens 2021; Allerbo 2025; Pezeshki et al. 2021; Brellmann et al. 2024; Nakkiran 2019a; Mei and Montanari 2019a; Heckel and Yilmaz 2020; Nakkiran et al. 2020; Belkin, Hsu, and Xu 2020; Yang and Suzuki 2024; Spiess, Imbens, and Venugopal 2023; Nakkiran et al. 2021; Y. S. Zhang 2024; Cherkassky and Lee 2024; Nakkiran 2019b. An even more exotic version of double descent, dubbed *triple descent* (or more), is discovered in Ascoli, Sagun, and Biroli 2020.

3. Theoretical foundation

For the need of analysing the phenomenon of double descent, we begin by listing and reviewing different foundational notions that describe the typical theory on the learning dynamics, and the learnability problem. Such ranges from statistical learning theory, as seen of Vapnik 1999; Hajek and Raginsky 2021; Mohri, Rostamizadeh, and Talwalkar 2012, and the more practical learning theory, as seen in James et al. 2009.

The setting of typical machine learning system comes in two-fold, either from the outer objective of the *observation space* $\{x_i, y_i\}_{i=1}^n$, denoted $\mathcal{S}$ for any 2-tuple of that form, or the general, oracle-induced view in which there exists the main form, or the oracle $c \in \mathcal{C}$, of which form the functional spaces. We would like to review these two outlooks.

3.1 General learning

Classically, we can see that the problem of double descent is not inherently a problem that can be explained only by considering model complexity and the error rate. As such, it requires a framework that consider multiple factors together, and interpreting different hypothesis. Nevertheless, classical theory does indeed point to certain factors or aspect that should be of concern. For example, we have seen plenty of analysis that indicates that *sample complexity* might be the biggest factor, considering the behaviour is intrinsically operable and perhaps observable in almost any model, except for certain model that is specialized of the data structure itself. Even by then, the error still can be obtained given large enough model disparity to the concept class. There are attempts to, indeed, analyze double descent from a VC perspective (Lee and Cherkassky 2022), however, the paper falls into the same pitfall as others in their interpretation and analysis.

Nevertheless, even though we have said plenty of the disadvantage of the classical theory in terms of statistical learning, their

insight is valuable in determining the possibility of dynamic in question. Thus of the foregoing sections about the setting itself.

The learning problem is then formulated as followed. Given the machine learning model expressed a hypothesis h of the hypothesis class $\mathcal{H}$, the learning theory aims for creating a procedure to learn either elements of the concept class $\mathcal{C}$ of all concepts $c : \mathcal{X} \rightarrow \mathcal{Y}$, or the function class $\mathcal{F}$ of all functions $f : \mathcal{X} \rightarrow \mathcal{Y}$, which is usually set to $\{0, 1\}$ or $[0, 1]$. This distinction is trivial, hence, if it is clear, we will talk about the concept class $\mathcal{C}$ only as representative form. The learner $\mathcal{L}(h)$ consider the set of possible hypothesis $\mathcal{H}$, in which might not coincide with $\mathcal{C}$. It receives a partial image of sample $S = (x_1, \dots, x_2, \dots, x_n)$ drawn i.i.d. according to $\mathcal{D}$ as well as the label $(c(x_1), \dots, c(x_n))$. This constitutes the dataset $\mathcal{S}$, which are based on specific concept $c \in \mathcal{C}$ for the hypothesis to learn. The task is then to use (or *extract*) meaningful information to select a hypothesis $h_S \in \mathcal{H}$ that accurately mimic c , with marginal error Θ . The notation h_S stands for all hypothesis that can be inferred from the range of the dataset (the first argument, $\mathcal{X}$).

The marginal error Θ is considered of two parameters, the empirical error $\hat{R}(h)$ and the generalization error $R(h)$. We give the following definition of them.

Definition 3.1 (Empirical risk). *Given a hypothesis $h \in \mathcal{H}$, a target concept $c \in \mathcal{C}$, and a sample $S = (x_1, \dots, x_m)$. For some particular $\epsilon > 0$, the **empirical error** or **empirical risk** of h is defined by*

$$\hat{R}_S(h) = \mathbb{P}_{x \in S \sim \mathcal{D}} [\ell\{h(x), c(x)\} \geq \epsilon] = \frac{1}{m} \sum_{i=1}^m \ell\{h(x_i), c(x_i)\} \quad (1)$$

The generalization risk is then defined for the entirety of the sample set, assuming to be densely populated on the interval of specification. Hence, $x \sim \mathcal{D}$ returns infinitely many samples.

Definition 3.2 (Generalization risk). *Given a hypothesis $h \in \mathcal{H}$, a target concept $c \in \mathcal{C}$, and an underlying distribution $\mathcal{D}$ on $\mathcal{X}$. For some particular $\epsilon > 0$, the **generalization error** or **risk** of h is defined by*

$$\begin{aligned} R(h) &= \mathbb{P}_{x \sim \mathcal{D}} [\ell\{h(x), c(x)\} \geq \epsilon] \\ &= \mathbb{E}_{x \sim \mathcal{D}} [\ell\{h(x), c(x)\}] \\ &= \int_{x \in \mathcal{D}} \ell\{h(x), c(x)\} dP(x) \end{aligned} \quad (2)$$

In essence, the empirical risk measure the hypothesis to observation error, and the generalization risk captures the difference of the hypothesis to the actual concept. Notice that by doing this, we have implicitly assumed that the observations and the concept is not the same thing, under the majority of scenarios.

3.1.1 PAC-learning setup

The PAC-learning process (Shalev-Shwartz and Ben-David 2014; Mohri, Rostamizadeh, and Talwalkar 2012; Hajek and Raginsky 2021) connects three complexity measure - the sample space complexity, representation complexity, and optimization complexity, to the evaluation of the statistical learning process. It refers to the categorization of various learning procedure that satisfies the Probably Approximately Correct criteria in its name, as such being followed.

We assume no structure of h and c . They can be functions, partial functions, relations, complex algorithms, or others. In a typically learning setting, we also have the argument of *preliminary knowledge*, presented in literatures of the term *inductive bias*. For example, if the concept class $\mathcal{C}$ is assumed, then we say the setting is **model-specific**. If there exists no hard assumption on $\mathcal{C}$, then the learning setting is said to be **model-free**. Then, the PAC-learning set for model-specific setting is defined as followed.

Definition 3.3 (PAC-learning). *A concept class $\mathcal{C}$ is said to be **PAC-learnable** if there exists an algorithm $\mathcal{A}$ and a polynomial function $\text{poly}(\cdot, \cdot, \cdot, \cdot)$ of 4-argument such that for any $\epsilon > 0$, $\delta > 0$, for all distribution $\mathcal{D}$ on $\mathcal{X}$ and for any target concept $c \in \mathcal{C}$, the following holds for any sample size $m \geq \text{poly}(1/\epsilon, 1/\delta, n, \text{size}(c))$:*

$$\Pr_{S \sim \mathcal{D}^m} [R(h(S)) \leq \epsilon] \geq 1 - \delta$$

for a given error measure $R(h_S)$. If $\mathcal{A}$ further runs in

$$\text{poly}(1/\epsilon, 1/\delta, n, \text{size}(c))$$

, then $\mathcal{C}$ is said to be **efficiently PAC-learnable**. When such algorithm exists, we call $\mathcal{A}$ a **PAC-learning algorithm**.

This can be easily extended into the model-free case, and further on, which is left in the appendix section. The reason for the argument $\text{size}(c)$ is as followed. Consider a class of concepts defined by the satisfying assignments of certain formulae. A concept from this class that satisfies such formulae, can be represented by a formula f , a truth table, or any given formulae that is tautologically equivalent formulae f' to f . PAC-learning bound argues in the sense of *computational complexities*, and *space complexities*. Specifically, the term $\text{poly}(1/\epsilon, 1/\delta, n, \text{size}(c))$ hopes to bound the required learning process to polynomial time, specified by 4 parameters. Here, n is the *input dimension*, $\text{size}(c)$ is the encoding complexity of the target concept, $\epsilon > 0$ is the *accuracy bound*, and δ is the *confidence bound* - the probability that h fails.

PAC-learning procedure gives, by assumption of a **consistent learner**, two different generalization bounds for two different cases. We first define the notion of a consistent learning algorithm, or consistent learner, for a concept class $\mathcal{C}$ and hypothesis h .

Definition 3.4 (Consistent Learner). *We say that an algorithm is a **consistent learner** for a concept class $\mathcal{C}$ using hypothesis class $\mathcal{H}$, if for all n , for all $c \in \mathcal{C}_n$ and for all m , given*

$$S = \{(x_1, c(x_1)), (x_2, c(x_2)), \dots, (x_m, c(x_m))\}$$

find an alternative reasonable form of such loss landscape. Suppose of a regression problem to construct a hypothesis function $f(x)$ from $(x_1, y_1, \dots, x_N, y_N)$ for the purpose of generalization – that is, predicting unseen variational values for different pair $(x_j, ?)$ such that $? = y_j + \epsilon$ for a conceivable implicit error. To be explicit about the relation of this problem, or f on the given data $\mathcal{D} = \{(x_i, y_i) \mid i \leq N\}$, denote $f(x; \mathcal{D})$ instead of f , the natural mean-square measure as a predictor is:

$$\mathcal{M}(f, y) = \mathbb{E} \left[((y - f(x; \mathcal{D})))^2 \mid x, \mathcal{D} \right] \quad (5)$$

for $\mathbb{E}[\cdot]$ the expectation wrt to a distribution P . Decomposing the right-hand side, we have:

$$\begin{aligned} \mathcal{M}(f, y) &= \mathbb{E} \left[((y - f(x; \mathcal{D})))^2 \mid x, \mathcal{D} \right] \\ &= \mathbb{E} \left[(y - \mathbb{E}[y \mid x])^2 \mid x, \mathcal{D} \right] + (f(x; \mathcal{D}) - \mathbb{E}[y \mid x])^2 \end{aligned} \quad (6)$$

Here, $\mathbb{E}[(y - \mathbb{E}[y \mid x])^2 \mid x, \mathcal{D}]$ does not depend on $\mathcal{D}$, but simply the statistical variance of y given x . The term $(f(x; \mathcal{D}) - \mathbb{E}[y \mid x])^2$ is considered a natural measure of effectiveness on $\mathbb{R}^n$ as a singular predictor of y . Now, for $\mathbb{E}_{\mathcal{D}}[(f(x; \mathcal{D}) - \mathbb{E}[y \mid x])^2]$ which depends on the training set $\mathcal{D}$ in its computation, is decomposed into the form of *bias-variance decomposition* terms, by derivation:

$$\begin{aligned} &\mathbb{E}_{\mathcal{D}} \left[(f(x; \mathcal{D}) - \mathbb{E}[y \mid x])^2 \right] \\ &= \underbrace{\left\{ \mathbb{E}_{\mathcal{D}}[f(x; \mathcal{D})] - \mathbb{E}[y \mid x] \right\}^2}_{\text{bias term}} \\ &+ \underbrace{\mathbb{E}_{\mathcal{D}} \left\{ (f(x; \mathcal{D}) - \mathbb{E}_{\mathcal{D}}[f(x; \mathcal{D})])^2 \right\}}_{\text{variance term}} \end{aligned} \quad (7)$$

We summarize this in the following statement.

Theorem 4.1 (Bias-variance decomposition). *Suppose the model $f(x; \mathcal{D})$ for the data $\mathcal{D} = (x_i, y_i)$ and its parameter x is defined. For y_i of the target concept's responses y , and consider a regression problem with the loss measure $\mathcal{M}(f, y)$ of mean squared risk, the following statement is true:*

$$\begin{aligned} \mathbb{E}[\mathcal{M}(f, y)] &= \mathcal{B}(f, y) + \mathcal{V}(f, y) \\ &+ \mathbb{E} \left[\mathbb{E} \left[((y - f(x; \mathcal{D})))^2 \mid x, \mathcal{D} \right] \right] \end{aligned} \quad (8)$$

for $\mathbb{E}[\cdot \mid x, \mathcal{D}]$ any expression with dependencies on x and $\mathcal{D}$. The bias and variance term is subsequently expressed by

$$\mathcal{B}(f, y) = \underbrace{\left\{ \mathbb{E}_{\mathcal{D}}[f(x; \mathcal{D})] - \mathbb{E}[y \mid x] \right\}^2}_{\text{bias}}, \quad (9)$$

$$\mathcal{V}(f, y) = \underbrace{\mathbb{E}_{\mathcal{D}} \left\{ (f(x; \mathcal{D}) - \mathbb{E}_{\mathcal{D}}[f(x; \mathcal{D})])^2 \right\}}_{\text{variance}} \quad (10)$$

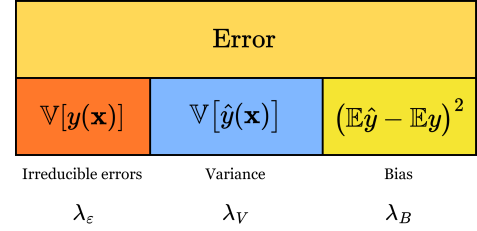


Figure 2. Decomposition of the error term into 3 parts. Respectively, the irreducible error $\mathbb{V}[y(\mathbf{x})]$, the variance (blue) and the bias (dark yellow). All of them are assumed to take up 100% of the error observed in such composition. The proportion is included using the three coefficient $\lambda_{\epsilon}, \lambda_V, \lambda_B$ where $\lambda_{\epsilon} + \lambda_V + \lambda_B = 1$.

The above decomposition principle is often expressed into a form where there exists the intrinsic noise Brown and Ali 2024:

$$\begin{aligned} \mathbb{E}_{\mathcal{D}} \left[\mathbb{E}_{xy} (y - \hat{f}(x))^2 \right] &= \mathbb{E}_x \left[\left(y^* - \mathbb{E}_{\mathcal{D}}[\hat{f}(x)] \right)^2 \right] \\ &+ \mathbb{E}_x \left[\mathbb{E}_{\mathcal{D}} \left(\hat{f}(x) - \mathbb{E}_{\mathcal{D}}[\hat{f}(x)] \right)^2 \right] \\ &+ \mathbb{E}_{xy} \left[(y - y^*)^2 \right] \end{aligned} \quad (11)$$

In all, almost all representation of bias-variance decomposition are the same. For simplicity of notation, we adopt the similar form in the standard case of the paper Adlam and Pennington 2020:

$$\mathbb{E}[\hat{y}(\mathbf{x}) - y(\mathbf{x})]^2 = (\mathbb{E}\hat{y}(\mathbf{x}) - \mathbb{E}y(\mathbf{x}))^2 + \mathbb{V}[\hat{y}(\mathbf{x})] + \mathbb{V}[y(\mathbf{x})] \quad (12)$$

A main common theme of criticism toward bias-variance trade-off is the fact that the decomposition is much more general, and intrinsic for the class of *mean squared loss*. However, when considering the naturalness of an error measure and then its direct applicant, mean square error on real space of sufficient support and measure comes of as a very natural choice of an error measure, at least in consideration of the regression setting; for that, we then can also define classification as another top-layer above a regression's similar real continuous space, i.e. a decision layer. Furthermore, it can also be shown Brown and Ali 2024; Pfau 2013 that it also holds for the class of Bregman divergence measure.

What does the decomposition say about model selection principle in general? In general, bias-variance is typically presented of its decomposition only, and the asymptotic prediction of its behaviours. In fact, one of the reason that it became the rule-of-thumb for ML practitioner, as well as generally statistical learning (Lafon and Thomas 2024 provides a quite rigorous treatment of bias-variance tradeoff in the section on statistical learning theory) solidify the trade-off as a particular model selection principle. Generally, this tradeoff can be summarized as followed:

Theorem 4.2 (Bias-variance tradeoff). *For the expected loss of any given hypothesis h , the bias $\mathcal{B}(f, y)$ and variance $\mathcal{V}(f, y)$ is inversely proportional, that is, $\mathcal{B}(f, y) \propto \lambda^{-1} \mathcal{V}(f, y)$ for some proportionality λ that may or may not be constant. In the most general case possible, $\lambda = -1$ on the entire error range.*

The tradeoff is then of inverse proportionality. Indeed, statistically, we have such tradeoff on a statistical framework in a more concrete sense. For the bias to increase, variance will increase, of which the criterion is inverse – we would like to have more bias but lower variance, and vice versa, according to such theory.

4.1.2 Expansion and variations

The bias-variance decomposition and tradeoff is not general. Indeed, works, by Geman, Bienenstock, and Doursat 1992; Sharma and Aiken 2014; P. M. Domingos 2000a; Adlam and Pennington 2020; Yang et al. 2020 and more all attempted to reconstruct and reformulate bias-variance, and it did leave out a picture of uncertainty regarding the concept. Specifically, there are certainly different way to do the bias-variance decomposition, as classified in Adlam and Pennington 2020 paper. Some of this are particularly specialized in *neural network* training, basing on the main structure of modern machine learning in neural network formalism.

1. **Semiclassical approach:** The semiclassical approach is fairly simple. We can rewrite to another variation of the classical approach to be

$$\begin{aligned} \mathbb{E}[\ell(h)] &= \mathbb{E}_x \mathbb{E}_\epsilon [\hat{y}(x) - y(x)]^2 \\ &= \mathbb{E}_x (\mathbb{E}_\epsilon [\hat{y}(x)] - y(x))^2 + \mathbb{E}_x \mathbb{V}_\epsilon [\hat{y}(x)] \end{aligned} \quad (13)$$

and instead, consider the *structural conformation* of the decomposition, or rather, taking the expectation within $\theta \in \Theta$ of the structural description of $h \in \mathcal{H}$. Then, we shall have

$$\begin{aligned} \mathbb{E}_{\text{SC}}[\ell(h)] &= \mathbb{E}_x \mathbb{E}_\theta \mathbb{E}_\epsilon [\hat{y}(x) - y(x)]^2 \\ &= \mathbb{E}_x \mathbb{E}_\theta (\mathbb{E}_\epsilon [\hat{y}(x)] - y(x))^2 + \mathbb{E}_x \mathbb{E}_\theta \mathbb{V}_\epsilon [\hat{y}(x)] \end{aligned} \quad (14)$$

Do note that the order matters, since the expectation can be constant of a given ordering. And expected value over constant is just the constant.

2. **Symmetric decomposition:** This approach is fairly difficult, we refer to the paper itself of its own justification on the decomposition. The symmetric decomposition relies on the shape of the input, $\mathcal{X} = \{P, D, \epsilon\}$, of which you can remove the ϵ term and retain some decomposition on P and D per distribution.

4.1.3 Generalized decomposition

According to P. Domingos 2000b, 2000a, the bias-variance decomposition can usually be treated as being controlled by three factor error, with their respective coefficients $\lambda = (\lambda_1, \lambda_2, \lambda_3)$

for such:

$$\begin{aligned} \mathbb{E}_{\mathcal{D}} [d(f, \mathbb{E}[y | x])] &= \lambda_1 \text{Bias}(f, y) + \lambda_2 \text{Var}(f, y) + \lambda_3 \epsilon(\mathcal{D}) \\ &= \lambda_1 \underbrace{\{\mathbb{E}_{\mathcal{D}}[f(x; \mathcal{D})], \mathbb{E}[y | x]\}}_{\text{bias term}} \\ &\quad + \lambda_2 \underbrace{\mathbb{E}_{\mathcal{D}} \{ (f(x; \mathcal{D}), \mathbb{E}_{\mathcal{D}}[f(x; \mathcal{D})]) \}}_{\text{variance term}} \\ &\quad + \underbrace{\lambda_3 \epsilon}_{\text{irreducible error}} \end{aligned}$$

where $\epsilon(\mathcal{D})$ is the irreducible error of the system, depends (theoretically) on the intrinsic imperfection of the dataset $\mathcal{D}$. This method of which the bias-variance-irreducible error are ‘fit’ into the total error by three coefficients $\lambda_1, \lambda_2, \lambda_3$ instead. This does not only scale up the B-V-IE triplet, but also leaves out the inexplorable gap between the scaling, and the actual normal measure.

4.1.4 Bias-variance algorithm

4.1.5 Functional approximation algorithm

Usually, the optimization problem is evaluated empirically of its bias-variance metric. That is, we test singular models successively, and then choose the generally best — usually within bias-variance tradeoff. This can be expressed as a function for theorem 4.2.

4.1.6 Approximation-estimation tradeoff

Another closely related notion to bias-variance is the concept of approximation-estimation tradeoff. We refer to Mohri, Rostamizadeh, and Talwalkar 2012; Lafon and Thomas 2024 for some analysis and mentions.

Let $\mathcal{H}$ be a family of functions mapping $\mathcal{X} \rightarrow \{1, -1\}$. This is the particular case of **binary classification**, in which can be straightforwardly extended to different tasks and loss functions. The *excess error* of a hypothesis $h \in \mathcal{H}$, is the difference between its error $R(h)$ and the Bayes error R^* . This can be decomposed to be the following:

$$R(h) - R^* = \left(R(h) - \inf_{h \in \mathcal{H}} R(h) \right) + \left(\inf_{h \in \mathcal{H}} R(h) - R^* \right) \quad (15)$$

The first bracket contains the **estimation error**, and the second bracket contains what is called the **approximation error**. The estimation error depends on the hypothesis h selected. It measures the error of h with respect to the infimum of the error achieved by hypotheses in $\mathcal{H}$, or that of the best-in-class hypothesis h^* when that infimum is reached. The approximation error measures how well the Bayes error can be approximated using $\mathcal{H}$. It is a property of the hypothesis set $\mathcal{H}$, a measure of its richness.

Model selection consists of choosing $\mathcal{H}$ with a favourable trade-off between the approximation and the estimation error. However, in the most general case, this will be done, but not in

practice, as it requires the underlying distribution $\mathcal{D}$ to be known to determine R^* , which is not possible. In contrast, the estimation error can be bounded, or can be analysed, using particular metric and analysis. It is however, worth to note that bias-variance and approximation-estimation have a very complicated relationship, for example, in Brown and Ali 2024 shown that they are indeed not the same, and is in fact two different decomposition, where one is the others' component. To see this, let us compare the two of them in details.

4.2 Double descent

Nevertheless, of bias-variance and the analogous statistical learning theory concept, the target is the same. It is the dilemma of which is presented in **Occam's razor**, for choosing the sufficient model of good complexity, or bias, for tradeoff of its generalization ability, or variance. Then there must exist a sweet spot between the axis of bias and variance, since they are as exhibited above inversely proportional to each other. However, double descent seemingly broke the status quo, and insists on an interesting phenomenon - under the same setting, if we 'crank' the complexity high enough, we will then reach a point then called the **interpolation threshold**, such that the trend reverse and the error rate, instead of being theorized to go up, goes down to a certain line of lower bound.

The first identification of the double descent phenomena dated back to the paper of Belkin - Belkin et al. 2019a, in which the title is literally "reconciling" modern machine learning practice and the bias-variance tradeoff. In modern machine learning practice, or state-of-the-art developments, models are now bigger than ever. If to notice, we will see that currently models are inherently large, for example, a normal large language model will have from 900 millions (900M) to a few billions, for example 10 billions (10B) parameters. That is not taking into account the overall dynamics and structure of the model, which dictates the operating range and efficiency of the model itself. These model, based on the neural network architecture are somewhat trained to exactly fit (or interpolate) the data, almost certainly so that it turn from a prediction setting to an estimation setting. By statistical learning theory, this would be considered overfitting, and yet, they often obtain very high accuracy on test data.

The main finding that Belkin found is a pattern for how the apparent performance on unseen data depends on model capacity and the mechanism underlying the emergence of double descent. When function class capacity is below the "interpolation threshold", learned predictors exhibit the classical U -shaped curve from Figure 3. The 'modern' interpolating regime marks the opposite trend to the right, where the risk starts to decrease up to a lower bound, which then can be called the *optimal descent bound*.

The bottom of the U is achieved at the sweet spot which balances the fit to the training data and the susceptibility to over-fitting: to the left of the sweet spot, predictors are under-fit, and immediately to the right, predictors are over-fit.

When we increase the function class capacity high enough (e.g., by increasing the number of features or the size of the neural network architecture), the learned predictors achieve (near) perfect fits to the training data—i.e., interpolation. Although the learned predictors obtained at the interpolation threshold typically have high risk, we show that increasing the function class capacity beyond this point leads to decreasing risk, typically going below the risk achieved at the sweet spot in the "classical" regime. (Belkin et al. 2019a)

At this point, we would like to also clarify the term interpolation threshold in this case. Usually, under the pretext of overfitting and underfitting, interpolation threshold means the following:

Definition 4.1 (Interpolation threshold). *Assuming a typical supervised learning setting, for a hypothesis h within class structure $\mathcal{H}$, and a supposed concept c of the class $\mathcal{C}$. Suppose $[X, Y] \subset c \in \mathcal{C}$ being the dataset, or observable space of c on which h is tested on. Under the **empirical generalization learning** procedure, partitioning $[X, Y]$ into $[X, Y]_1$ and $[X, Y]_2$, for error evaluated on the first set, $\hat{R}_1(h)$ called training error, and the second set, $\hat{R}_2(h)$ called test error, with optimization applied on the first set, then evaluated statically on the second, the **interpolation threshold** with respect to a particular complexity measure of h , $\mathcal{M}(h)$, is the point $\mathcal{M}_0(h)$ where it is possible for $\hat{R}_2(h)/\hat{R}_1(h) \approx 1$, for $\hat{R}_1(h) \ll \epsilon$, $\hat{R}_2(h) \ll \epsilon$, with $\epsilon \geq 0$.*

This basically means that what is learnt, with respect to the model complexity that gauge *how much* and *what* the model can learn, perfectly fits the concept c itself, from the observation space on its own. We can say that, that specific sense, we gain the assumption that the observation space is *fully representative* of the concept class, and the concept class is then fully realized by the model. Such point where that is possible, or there then such behaviour is possible, is called the interpolation point. The definition relies on the statement 'where it is possible', hence it still, does not give us a particularly sound notion on to how and what constitute such possibility. The term empirical generalization learning comes from the fact that the procedure in practice of estimating and optimizing train-test-validation set in itself, is an imperfect empirical approximation of generalization testing. While the fact that we cannot gauge generalization error, in a purely blind black-box setting is settled (Mohri, Rostamizadeh, and Talwalkar 2012; Sugiyama 2015; Shalev-Shwartz and Ben-David 2014), the method is then to try to mimic such by using generalization-via-hidden observations. To facilitate this, deliberately, the empirical learning procedure is only dynamic during training session, and static throughout the whole testing and validation testing section, and further on.

By default, in general theory, such interpolating point is dangerous. Not only because it is inherently difficult to know exactly what the dataset constitute, whether it is actually representative of the concept that gives rise to it, or the structural

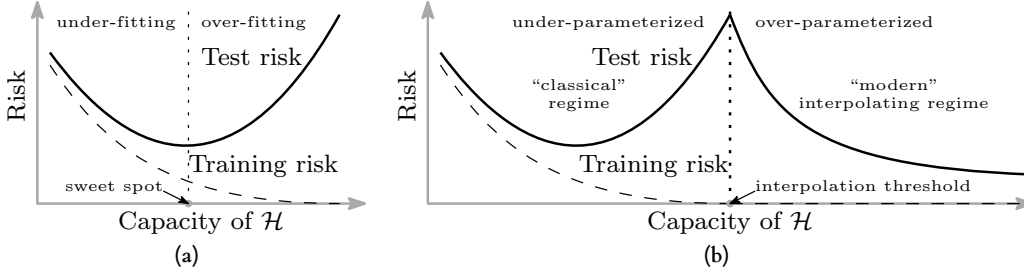


Figure 3. Curves for training risk (dashed line) and test risk (solid line). (a) The classical U-shaped risk curve arising from the bias-variance trade-off. (b) The double descent risk curve, which incorporates the U-shaped risk curve (i.e., the “classical” regime) together with the observed behaviour from using high capacity function classes (i.e., the “modern” interpolating regime), separated by the interpolation threshold. The predictors to the right of the interpolation threshold have zero training risk. Reproduced from Belkin et al. 2019a.

information, noises, and interferences in between. It is hence impossible, to a certain point, to truly know the effectiveness of the model beside interpolating the current dataset which forms the observable details.

Another prominent discovery to look at is Nakkiran et al. 2019, on the double descent of deep learning models. The paper serves as the first observation upon which double descent is observed on modern, deep learning theoretic models, especially complex one like Transformer or ResNet networks.

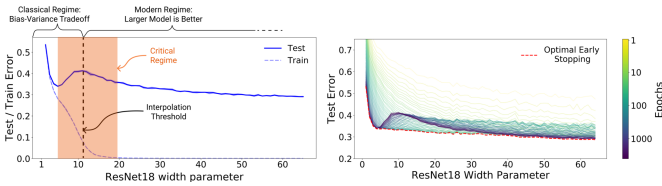


Figure 4. Left: Train and test error as a function of model size, for ResNet18s of varying width on CIFAR-10 with 15% label noise. Right: Test error, shown for varying train epochs. All models trained using Adam for 4K epochs. The largest model (width 64) corresponds to standard ResNet18. Resued from Nakkiran et al. 2019.

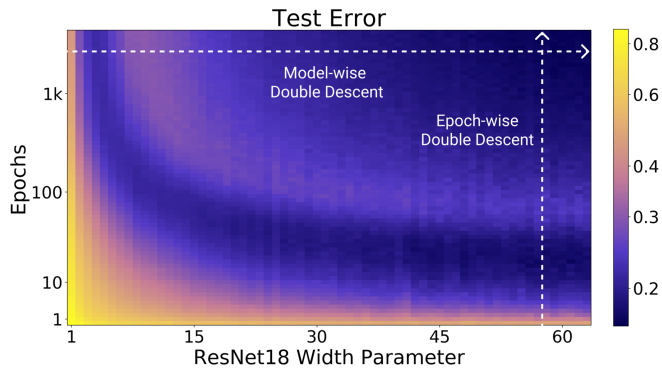


Figure 5. Test error as a function of model size and train epochs. The horizontal line corresponds to model-wise double descent-varying model size while training for as long as possible. The vertical line corresponds to epoch-wise double descent, with test error undergoing double-descent as train time increases. Resued from Nakkiran et al. 2019

They define *effective model complexity* of $\mathcal{T}$ (w.r.t. distribution $\mathcal{D}$) to be the maximum number of samples n on which $\mathcal{T}$

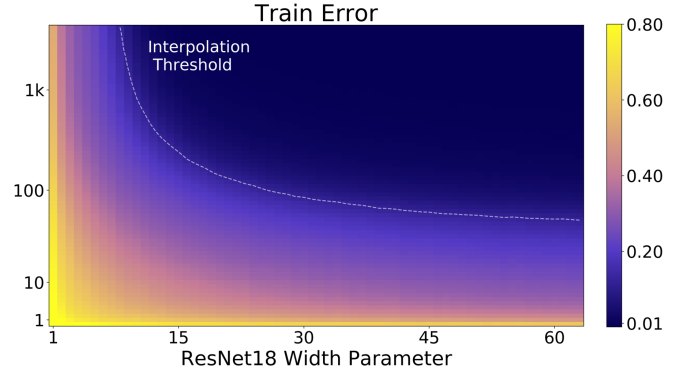


Figure 6. Train error of the corresponding models. All models are Resnet18s trained on CIFAR-10 with 15% label noise, data-augmentation, and Adam for up to 4K epochs. Reused from Nakkiran et al. 2019

achieves on average ≈ 0 training error. This is an entirely empirical definition, similarly per definition as VC-dimension.

Definition 4.2 (Effective Model Complexity). *The Effective Model Complexity (EMC) of a training procedure $\mathcal{T}$, with respect to distribution $\mathcal{D}$ and parameter $\epsilon > 0$, is defined as:*

$$\text{EMC}_{\mathcal{D},\epsilon}(\mathcal{T}) := \max \{n \mid \mathbb{E}_{S \sim \mathcal{D}^n} [\text{Error}_S(\mathcal{T}(S))] \leq \epsilon\}$$

where $\text{Error}_S(M)$ is the mean error of model M on train samples S .

Using this definition, their main hypothesis can then be stated as the following three-fold regions:

Hypothesis 4.1 (Generalized Double Descent hypothesis, informal). *For any natural data distribution $\mathcal{D}$, neural-network-based training procedure $\mathcal{T}$, and small $\epsilon > 0$, if we consider the task of predicting labels based on n samples from $\mathcal{D}$ then:*

Under-parameterized regime. *If $\text{EMC}_{\mathcal{D},\epsilon}(\mathcal{T})$ is sufficiently smaller than n , any perturbation of $\mathcal{T}$ that increases its effective complexity will decrease the test error.*

Over-parameterized regime. *If $\text{EMC}_{\mathcal{D},\epsilon}(\mathcal{T})$ is sufficiently larger than n , any perturbation of $\mathcal{T}$ that increases its effective complexity will decrease the test error.*

Critically parameterized regime. *If $\text{EMC}_{\mathcal{D},\epsilon}(\mathcal{T}) \approx n$, then a perturbation of $\mathcal{T}$ that increases its effective complexity might decrease or increase the test error.*

It is to notice that this definition is also only observational. By that, it only outlines the specific region of interest where the paradigm shift is identified through experimental results. It has no effective predictive power, or rather descriptive power more than setting up hypothesis with respect to the effective model complexity, and some arbitrary perturbation. Nakkiran et al. 2019 also noticed this difficulty in providing a theoretical definition and theorem regarding such hypothesis, as said in their manuscript. The existence of the critical regime is also not defined.

The behaviour itself has been particularly investigated, in certain settings. For example, before Belkin et al. 2019a, Advani and Saxe 2017 investigated the generalization error in neural networks of high-dimensional measure. Of such, various behaviours that bear similarity to double descent can be observed. Belkin, Ma, and Mandal 2018 expanded on his previous research on the particular subset of kernel learning models, providing a theoretical analysis of specifically the Laplacian kernel on standard neural network. Mei and Montanari 2020 investigated random feature regression in such regard, and also found similar results.

The fact that double descent is not well-defined itself is a problem on its own. Up to the author's knowledge and of myself, there has been no effective definition or given description that outlines specifically how double descent can be structured. Instability in reproducing experiments and the effective range of the phenomena remains a question on its own.

5. Current researches

Ever since Belkin et al. 2019a, despite the general chaotic view of double descent that seems to hold up pretty well, many researches have been conducted throughout, trying to observe or explain it, as we briefly mentioned Nakkiran et al. 2019 as one of such main paper. Before attempting to do anything, and to further contribute with hypotheses forward, we will have to resolve to review previous coursework on such matter. Most of those works include some on theoretical analysis, some on dissection of optimization method then speculation of possible hindsight, some on experiments and observable effects, and so on.

As such, the first thing to do is to contrive the consensus on 'what' constitutes double descent, at least per observable results, by delivering a conjecture.

Conjecture 5.1. *On the landscape of double descent, model complexity versus generalization error, there exists a fundamental point $\mathcal{M}_0$ of model complexity axis, where $dR(h)/d\mathcal{M} < 0$ for $\mathcal{M} > \mathcal{M}_0$, which is the interpolation point.*

While this is not perfect, it provides us with a general idea on where to look for double descent. With this, we are ready to follow up the literatures in provision of said questions. Let us also recite what happens within the setting of many double descent results.

5.1 Discovery of double descent

While double descent is discovered by Belkin et al. 2019a and further on modern deep learning landscape of Nakkiran et al. 2019, it remains that the conclusion is not reached if there exists some fundamental models that would not have double descent, the consistency of double descent through different models entirely and so on. This is mainly because of a particular 'shock effect', since before double descent, bias-variance holds the candle as the main principle of choosing and optimizing models to its optimal saddle point. Suddenly by then we observe double descent, which introduces the interpolating threshold $\mathcal{M}_0$, that breaks this principle. There is then the open question on how far-reaching is double descent in consideration of different architectures, and this is reflected in several papers above. As of now, the main structures that have been verified includes

- **Classical models** Yang and Suzuki 2024; Nakkiran 2019a; Schaeffer et al. 2023; Lafon and Thomas 2024 (linear regression and classical polynomial), Li and Sonthalia 2024 (linear regression upon least square), Liu, Liao, and Suykens 2021, Y. S. Zhang 2024; Allerbo 2025 (kernel methods/regressions), Brellmann et al. 2024 (Classical MSBE-LSTD^a optimization), Harzli et al. 2023 (Gaussian process). Some papers also have experiments on Random Fourier Feature Networks, and so on, but not substantial (including Belkin et al. 2019a paper).
- **Deep neural networks:** Luo and Dong 2023; Yang and Suzuki 2024; Pezeshki et al. 2021 (CNN, VGG, ResNet, DenseNet), Nakkiran et al. 2019 (Transformer), Shi et al. 2024 (GNN).

Most of the experiments are concerned of seeing and analysing double descent, basic analysis and experimental observations, test case scenarios for different models and setting and so on. All of the above models are then considered to be, concretely, reported of observing double descent, through various testing scenario on them. Review papers and literatures can be found as Schaeffer et al. 2023; Hu 2021, and Shi et al. 2024 on specifically, graph neural network.

5.2 Theoretical attacks

Because of its intrinsic nature, where double descent breaks the long-standing principle of learning selection, theoretical resolutions has been employed to tackle such problem. Indeed, for example, Cherkassky and Lee 2024 and Lee and Cherkassky 2022 aim toward such direction by understanding the behaviours of double descent using the classical VC-theory approach. However, such papers only introduce, or relate double descent to VC-theory via phenomenological approach – for example, reducing the VC-dimension while fixing or reducing training error, or by examining a potential proxy between the VC-bounds and potential factors encapsulated in the quantities involved. They however, do raise voices of certain inherent problems lies within the theory of learning and in general, learning models.

Beyond such, certain discovery and inquiries help in having

a. Mean-Squared Bellman Error and Least-Squared Temporal Difference

a clearer picture on the behaviour of double descent through various structures. For example, Hastie et al. 2019, under pre-text of studying high-dimensional ridgeless (ℓ_2 norm) interpolation, analyse the case of *isotropic features*, *correlated features*, misspecified model (feature observation inhibited), on ridgeless interpolation of linear model, and soon as nonlinear model and latent space model. While strong, their results are restricted to a specific class of assumptions and structure, and is doubtful to be extended further. Allerbo 2025 investigated and find a particular double descent trace when changing kernel during training, specified in the term of the minimum allowed bandwidth. The kernel bandwidth $\sigma \geq 0$ of a specific kernel filter determines the information smoothing, hence implies the complexity abides by the kernel structure, as well as determine the smoothness of information indicted by the term.

The sentiments that are shared in between researchers on the topic of double descent vary, but have some common roots, most of them coming from the difficulty to fully assess double descent and its factor components. While double descent concerns the evaluation of *generalization (test) error*, and the *model complexity* notions, the error in question is often implicitly defined or loosely considered in one way or another, and model complexity has no general consensus on its definition. In fact, this points out a very potential, if not already mismatch in the theoretical development. While optimizations and phenomenological aberration of models, as well as building and constructing models at will are well-developed, the definitions on which they are defined on is not fully understood, if not undefined. As of now, literatures on model complexities like al., Hu et al. 2021; Janik and Witaszczyk 2021; Truong 2025; Luo, Wang, and Huang 2024 introduce their variations or a copula of different definitions altogether, where some are concerned with its effective complexity, some are concerned with its expressive or structural complexity with different ways to define the model underneath, from parameters to units, to random standard variables (Neal et al. 2018). There is also the rift in consideration between classical theory, of which mainly is for classical models, and modern deep neural network structures of which do not conform to such analysis.

5.3 High-dimensional ridgeless interpolation - Hastie et al. 2019

Aligning with our previous experiments, Hastie et al. 2019 conducted investigation into the class of ridgeless approximation criterion on linear-nonlinear classes disparity. The investigation itself is thorough in parameter-term. Specifically, they investigated the case of which we have **isotropic features**, where asymptotic risk depends on the weight w only through its norm $\|w\|_2$; **anisotropic features**, where the model risk depends on the geometry of the pair (Σ, w) for $\Sigma = \text{Cov}(x_i)$ on the dataset $(y_i, x_i)_{i=1}^n \in \mathcal{S}$; and the case of nonlinear model, where the model itself is tested with added structure - x_i is obtained by passing $z_i \sim \mathcal{N}(0, I_d)$ through a one-layer neural network with random first layer weights.

5.4 Sample-wise double descent - Nakkiran 2019a

5.5 Multiscale feature learning dynamics - Pezeshki et al. 2021

5.6 Reinforcement learning and double descent - Brellmann et al. 2024

5.7 Out-of-Distribution Detection - Ammar et al. 2025

5.8 Gaussian processes and neural network - Harzli et al. 2023

5.9 Double descent on GNN - Shi et al. 2024

5.10 Weak feature double descent - Belkin, Hsu, and Xu 2020

5.11 Grokking, Emergence, and Double Descent - Huang et al. 2025

5.12 Variations on analysis

Many sub-variations and analysis alongside major substantiated discoveries listed above can help in analysis. Some of them are toward linking different existing dynamics to large neural network in non-linear fashion, or generalizing approximation of linear network toward nonlinear systems.

5.12.1 Dropout and double descent - Yang and Suzuki 2024

Dropout is a relatively well-established regularization technique for training deep neural networks, usually by exclusion of different neurons during processes, as seen i Srivastava et al. 2014 to control overfitting. Accordingly, Yang and Suzuki 2024 predicted or stated that regularization techniques can nullify double descent, and thus also dropout. The model considered is linear regression model, for p covariates $x \in \mathbb{R}^p$ and response or output $y \in \mathbb{R}$ are related by

$$y = x^\top \beta_0 + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2) \quad (16)$$

The situated dropout is as followed. Given training data $z^n = \{(x_i, y_i)\}_{i=1}^n$, for $X = [x_1, \dots, x_n]^\top$ and $y = [y_1, \dots, y_n]^\top$, the estimation of weight β by $\hat{\beta}(z^n)$ (model's weight), to minimizing the training error

$$\mathbb{E}_{r \sim \text{Ber}} \|y - (r \odot X)\beta\|_2^2 = \|y - \gamma X\beta\|_2^2 + (1 - \gamma)\gamma \|\Gamma\beta\|_2^2 \quad (17)$$

for $\odot$ denoting the element-wise product on L^2 norm space, each element of $R \in \mathbb{R}^{n \times p}$ takes one and zero with probabilities γ and $1 - \gamma$, for the distribution $\text{Ber}(\gamma)$ corresponding to such, and $\Gamma = \text{diag}(X^\top X)^{1/2}$. The final description gives the form of the effective dropout objective. Originally from the paper, they consider it exactly of a Tikhonov regularization method. Set $\beta' = \gamma\beta$, then

$$\begin{aligned} \|y - X\beta'\|_2^2 + \frac{1 - \gamma}{\gamma} \|\Gamma\beta'\|_2^2, \\ \hat{\beta}_{n\gamma} = \left(X^\top X + \frac{1 - \gamma}{\gamma} \Gamma^\top \Gamma \right)^{-1} X^\top y \end{aligned} \quad (18)$$

minimization. While experiments regarding such regularization works in such domain and was illustrated as to be reasonably effective, fallacies in claims and insight remains, especially in the nullification of double descent. The problem here is that, any given amount of dropout is, accordingly to the author, similar to Tikhonov or ridge regularization on structural complexity. As explained in the c' optimization of c -space in preliminary sections, such induces a different, much smaller

system in which the additional structural attractor points toward the region of low regularizing complexity throughout the domain. Such superposition between the $\{\lambda r(h)\}(\mathbb{R})$ landscape and the classical loss $\hat{R}$ only morph the dynamic setting to a different region, and thereby bypass double descent in such case for absolute non-overfitting convergence to $\min(\hat{R}_S(h))$. This works well on well-behaved and loosely stable case, but in a lot of scenarios, such cannot be guaranteed, especially for pathological systems. Furthermore, Speaking in terms of the *inductive bias*, mismatch between model and effective regularization becomes a problem, as there exists no effective complexity measure for the regularization term $r(h)$. Some are ill-conditioned, for example, $r(h) = \sum_j w_{ij}^2$ for square terms ij -indexed weights only take into account the reasonable weight system, without the semantic itself, or meaningful weights. A forced dropout might potentially in certain less ideal cases, be more destructive, and introduces double descent back again, until the stochastic process shifts meaningful weight to a more resource constrained system. This means we have to induce, or at least formalize if stochastic process can recover meaningful semantic after dropout randomly. We can test this somewhat, by introducing a small toy test on the proposed dropout estimator:

Theorem 5.1 (Expected test error, Yang and Suzuki 2024).

Let $\alpha = C < \frac{1}{(1+\sqrt{p/n})^2}$, the expected test error is

$$f(\alpha) = \left\{ 1 - a\alpha + 2\alpha^2 \frac{p}{n} \right\} \|\beta_0\|^2 + \sigma^2 \alpha^2 (\alpha + 1) \frac{p}{n} + O\left(\frac{1}{n^2}\right) \quad (19)$$

with $\alpha = \gamma/(1 - \gamma)$.

While the theorem is proved, we then must know if such theorem correctly correlates to even small dynamic systems.

6. Hypothesis

In learning and researching of the progress and treatment of double descent, several hypotheses, most of the time observations and insights such, is formed. Hence, there are several hypotheses for what to look out for, as well as the formulation needed to interpret certain phenomena and observations. This ranges from topics of re-evaluating the statistical learning theory, estimation and refactor of assumptions (or *inductive bias* when designing models), to different interpretations and representations that can better explain, model, or gives certain insights.

6.0.1 Stability of bias-variance measure

Stability of bias-variance measure is questionable, as it is in literature (P. M. Domingos 2000a), loosely defined by defining

a 3-dimensional vector $\lambda = (\lambda_1, \lambda_2, \lambda_3)$ for such:

$$\begin{aligned} & \mathbb{E}_{\mathcal{D}} [d(f, \mathbb{E}[\gamma | x])] \\ &= \lambda_1 \text{Bias}(f, \gamma) + \lambda_2 \text{Var}(f, \gamma) + \lambda_3 \epsilon(\mathcal{D}) \\ &= \lambda_1 \underbrace{\{\mathbb{E}_{\mathcal{D}}[f(x; \mathcal{D})], \mathbb{E}[\gamma | x]\}}_{\text{bias term}} \\ & \quad + \lambda_2 \underbrace{\mathbb{E}_{\mathcal{D}} \{ (f(x; \mathcal{D}) - \mathbb{E}_{\mathcal{D}}[f(x; \mathcal{D})])^2 \}}_{\text{variance term}} \\ & \quad + \underbrace{\lambda_3 \epsilon}_{\text{irreducible error}} \end{aligned}$$

where $\epsilon(\mathcal{D})$ is the irreducible error of the system, depends (theoretically) on the intrinsic imperfection of the dataset $\mathcal{D}$. Assessment on stability of such decomposition is recommended.

6.0.2 Alternative measures

We believe that the bias-variance decomposition is a poor estimation and measure for controlling, validating and choosing model from, at least from the perspective of analysing missing details. Specifically, bias and variance term, as well as their supposed expression of decomposition in the standard loss measure of model effectiveness is not representative, nor expressive of the actual, underlying notion of *model complexity* $C(f)$ and its effectiveness. Furthermore, bias-variance decomposition is not straight-forward of others type of loss functions, except only for some classes of loss function (loss on Bregman divergences and exponential class). This sentiment is shared in Brown and Ali 2024.

6.0.3 Inductive bias

Inductive bias obviously plays a role, as mentioned by some literature^b, though we don't know exactly how the inductive bias can be formed to play any role in the central dynamic of the model operating process. Certain assumption, for example, the *convexity of the loss function* might actually affect the model as it is, and introduce unwanted patterns in the following process.

6.0.4 Randomness and probabilistic setting

Random initialization of parameters, as well as the initial conditions of the model initialization process might also be taken into account. Obviously, this is not a new insight, and is rather a very old one, however, this approach still has varied potential for interpreting double descent. Scaling up/down this effect might result in new insight, as well as inspecting the uncertainty and diffusion patterns in large-scale model deployment, in which from an end-to-end perspective requires several space transformation operations in successions.

b. Most notably, Lafon and Thomas 2024, of which raised particular point about the need of learning with inductive bias as a structural presumption. Additionally, the paper also includes an analysis on the inductive bias of gradient descent, though solutions and particular application on double descent explanation is doubtful.

6.0.5 Knowledge masking

The very ambiguous idea or treatment of statistical learning, as well as machine learning, is the idea of masked. Or, paraphrased, *hidden information* regarding the underlying process of estimation. Specifically, for example, we tend to assume things to not be able to grasp – either the probability $p_X(c)$ of the concept class, the distribution $\mathcal{D}$ specified to such input distribution, or the concept class $\mathcal{C}$, Markov properties, etc. We would like to formalize this notion as soon and as such in the most direct way, as it is one source of the many assumptions that we have in concession of theoretical results.

6.0.6 Intrinsic Data Factors

The effect that data has on the model should also be taken into account, as of fact. As far as we are concerned of, the shape, practical system that the dataset gives the setting, the factors running into the dataset, the choice of the dataset being sampled, selected in partitions, and other factors must also be analysed. Without such, we leave ‘gaps’ in which sensitive, potentially chaotic behaviours might occur frequently and would not be identified, or being classified into irreducible errors, static of the learning setting. As such, we might as well list a few potential points of interest.

1. **Data distribution:** the distribution of the dataset itself affects how the concept is represented in the space. If the representation space created by using the data is large, but sparse, then the condition would make for some behaviours to occur more easily. This is one degree of freedom of the system setting. If the dataset spread is low, yet the data is very volatile, then it will lead to inconclusive picture of the actual concept, hence introducing new behaviours on the now unknown region of the distribution. As such, the majority of those unknown spaces will treat every point there as a potential outlier instead.
2. **Data complexity:** The complexity of the data itself is of interest. In a controlled setting of response-inducing evaluation, identifying behaviours of the model with a glimpse on the data complexity – how the actual data works with the internal mechanics accompanied – is of specific interest, as it can affect some of the majorative effects and chaotic behaviours that occur in the learning process, especially stochastic process and probabilistic optimization patterns.
3. **Data partitions:** The partitioning of data in the previously mentioned sample partitioning section might give rise to previously observed behaviours, as well as explaining particular volatility when changing the test partitioning set. The ordering of the dataset also helps in such regard.

It is also worth to note that since we emphasize the use of dataset partitioning to replicate the empirical-generalization guarantee, it is also then indicative that such partitioning would have its separate way of calculating test error on itself.

Intuitively, it can be illustrated as followed. Suppose of given two datasets $\mathcal{S}_1, \mathcal{S}_2$, we can trace a pattern, if, for assumption, the partition is $k = 10$. If the algorithm $\mathcal{A}$ attempt empirical learning optimization specifically, and best optimize on indi-

vidual partition $\mathcal{S}_1^{(i)}$, then the path toward each and individual partition would potentially be very volatile, depends heavily on the partition randomization, and others factor leading to such. If the algorithm is justified in that sense, to ‘decelerate’ the optimization to a not so tight fit, the generalization would be much better. This is also the reasoning for the technique of **regularization**, and generalization task. However, again, such procedure cannot help but having volatile dataset itself as a factor of uncertainty. More specifically, the result might work for $\mathcal{S}_1$, but for another dataset $\mathcal{S}_2$, assuming no intersection, that is, $\mathcal{S}_1 \cap \mathcal{S}_2 = \emptyset$, then optimization based on one dataset would not be optimal in another, more so can be counterproductive. This is also used of the reasoning for **cross-validation** technique, of which aims to stabilize the partitioning. However, analytical analysis suggests a wider role of such, in the wider setting of learning theory, particularly in double descent later on.

7. Analysis on-claims

For understanding and analysing double descent and bias-variance trade-off, and furthermore in later section on identifying double descent, we would like to use test models specifically for exhibiting bias-variance, as well as testing hypothesis and forming theoretical conjectures. This is because of the troublesome nature of the phenomena itself, which proves difficult to analyse without sufficient experimental observations. Furthermore, specific analysis on these types of model might prove useful for generalization toward more difficult and complex structure.

Our major focus is on three types of model. First two are classical models and system, linear models on the real space $\mathbb{R}^n \rightarrow \mathbb{R}$. Second, is the type of **support vector machine** (SVM) models to highlight the analysis toward data-based fundamental analysis. The third one is the class and collection of different *neural network structures* (MLP, or variations). Our goal is then to expansively analyse, formalize, making conjectures and hypothesis and test them, to find, force double descent out, or identify it on the learning landscape. We note that structures of those models are not well-generalized, hence analysis from one might not apply or apply only in small conjunction. This approach is rather quite helpful in its own merit, as its choice and preference will gauge our parameters than we gauge the phenomena of the parameter it induces.

7.1 Setting: gradient-based optimization

In our usual experiment, we would most specifically, use a closely-regarded family of gradient-based algorithms, such as gradient descent (Cauchy 1847; Robbins and Monro 1951; Widrow and Hoff 1960; Hestenes and Stiefel 1952; Fletcher and Reeves 1964), mirror descent, natural descent, and so on (Nemirovski and Yudin 1983; Amari 1996, 1998; Yang and Amari 1998; Rattray, Saad, and Amari 1998; Gunasekar, Woodworth, and Srebro 2021; Luong et al. 2016; Xiao and Zhang 2014; Sohl-Dickstein 2012; Raskutti and Mukherjee 2013). While implementation and details of the specific al-

gorithm can be different, they all shares some of the same concepts of which making them possible for optimization in the specified iterative domain. The most general formulation possible is the following algorithm on convexity:

Algorithm 1: Gradient descent on general convex function

Data: $(x_i, y_i) \in \mathcal{S}$, η , $\nabla(\cdot)$ operator.

Result: Expected optimization to the minimal error on $\ell(\cdot, \cdot)$.

```

begin
   $x_1 \leftarrow$  randomly initialized points;
  for  $t \leftarrow 1$  to  $T$  do
     $x_{t+1} \leftarrow x_t - \eta \nabla f(x_t)$ 
  end
end
return  $\hat{x} = 1/T \sum_{t=1}^T x_t$ 

```

Nevertheless, one should be cautioned of the intrinsic assumption on which gradient descent of this type assumes. It assumes *convexity*, which can only be possible, if the landscape itself is supposedly, assumed to contain such global minima. Said global minima in a richer language can be described as the epistemic collapse of the learning structure, under the arbitrary optimization space. This is a *very serious assumption*, and should it fail, as seen in a lot of non-populated or general observation space, the entire procedure is invalid. It is also troublesome as to rely on the singular objective (numerically encoded) space in which it is trivial to realise that there are indeed, local minima. This scaling, with respect to both locality and global consideration, must be considered carefully, since *every single contribution* might distort the experiment in a very unexpected way.

7.2 Linear function class $h(u)$ - Lafon and Thomas 2024

Linear-linear scenario learning is one particularly well-studied case of representing double descent or not, usually through simplicity in analysis (of the similar ease of proof compared to binary classification). Models in this setting can be said to share the same *parameter space* with no additional structure or scaling mask (like in monomial basis). The model that is representative of this preliminary observation is from Lafon and Thomas 2024, specifically, linear regression (function approximation) with Gaussian noise, and of univariate case. Consider the family class $(\mathcal{H}_p)_{p \in \llbracket 1, d \rrbracket}$ of linear functions $h : \mathbb{R}^d \mapsto \mathbb{R}$ where exactly p components are non-zero ($1 \leq p \leq d$), we have the following model.

Definition 7.1 (Lafon and Thomas 2024). For $p \in \llbracket 1, d \rrbracket$, $\mathcal{H}_p$ is the set of functions $h : \mathbb{R}^d \mapsto \mathbb{R}$ of the form:

$$h(u) = u^T w, \quad \text{for } u \in \mathbb{R}^d$$

With $w \in \mathbb{R}^d$ having exactly p non-zero elements. The complexity here itself can be interpreted as the effective random initialization, trimming down data of hidden details instead - this is indicated

by the amount of information it observes given d dimension, and p -hypothesis dimension.^c

Using such model, they consider the setting of minimizing the customized mean square loss,

$$\min_{w \in \mathbb{R}^d} \frac{1}{2} \|Xw - Y\|^2 \rightarrow \min_{w \in \mathbb{R}^p} \frac{1}{2} \|X_{\sim p} w - \gamma\|^2 \quad (20)$$

of the sub-problem on p cardinality, where again p is the dimensionality or independent degree of freedom. This results in the following theorem that 'mimic' double descent on this linear regression model.

Theorem 7.1 (Double descent on linear regression - Lafon and Thomas 2024). Let $(x, \epsilon) \in \mathbb{R}^d \times \mathbb{R}$ independent random variables with $x \sim \mathcal{N}(0, I)$ and $\epsilon \sim \mathcal{N}(0, \sigma^2)$, and $w \in \mathbb{R}^d$. We assume that the response variable y is defined as $y = x^T w + \sigma \epsilon$. Let $(p, q) \in \llbracket 1, d \rrbracket^2$ such that $p + q = d$, $X_{\sim p}$ the randomly selected p columns sub-matrix of X . Defining $\hat{w} := \Phi_p((\hat{w}), [\hat{w}])$ with $(\hat{w}) = X_{\sim p}^+ \gamma$ and $[\hat{w}] = 0$.

The risk of the predictor associated to $\hat{w}$ is:

$$\begin{aligned} & \mathbb{E}[(y - x^T \hat{w})^2] \\ &= \begin{cases} (\|w_{\sim q}\|^2 + \sigma^2)(1 + \frac{p}{n-p-1}) & p \leq n-2 \\ +\infty & n-1 \leq p \leq n+1 \\ \|w_{\sim p}\|^2(1 - \frac{n}{p}) \\ \quad + (\|w_{\sim q}\|^2 + \sigma^2)(1 + \frac{n}{p-n-1}) & p \geq n+2 \end{cases} \end{aligned} \quad (21)$$

The point $p = n$ is representative of the interpolation threshold from underparameterization to overparameterization.

We present two parts - first are those fundamentals predicted by the theorem, and second are verifications of the theorem with our own version of testing. The first experimental setting is then conducted with fixed d , varying p for each varying n samples. Since this is a least square result, preliminary run is conducted using one-shot error-risk measure on least square closed-form approximation. Note that $p \leq d$, hence we are regarding the domain of the term underparameterized and equal parameterization setting. The result is presented in Figure 7

As we see from this particular experimental result, the theorem predicted a loose, left-peak absent, double descent curve estimation for a variety of test case with more and more samples in increment. There exist some fundamental conclusion that we have about the shift of the interpolation threshold upon to $p = n$, which is reflected in the piecewise prediction. However, this pattern breaks down for n higher than specific constant, for this case, starting from $n \approx 100$ to beyond. From here, at least from testing scenario, the test indicates that predictions suggest a breakdown of the double descent estimation.

^c. According to the *manifold hypothesis*, there are chances where the actual concept can be interpreted and mimicked by seeing that it lies on a lower-dimension manifold inside its actual expressive dimensional space.

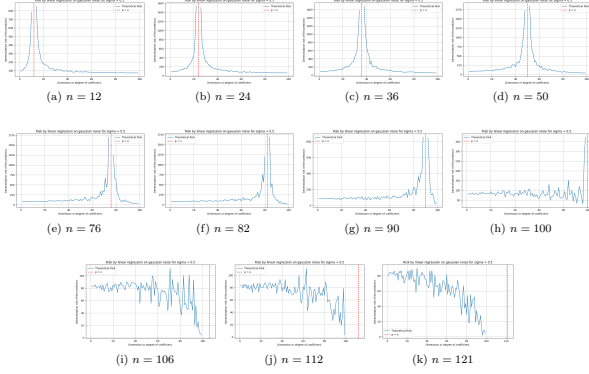


Figure 7. Theorem 7.1 behaviours on randomized setting. For this model, we have $d = 100$, $\sigma = 0.5$, and variational n . Full test cases are for $p = [1, 100]$, $n = \{12, 24, 36, 50, 76, 82, 90, 100, 106, 112, 121\}$ accordingly. The test function of the concept itself is the same linear model, but with different input only.

Beyond this, a simple semi-chaotic convergences is observed, from $n = 106, 112, 121$.

Interestingly, verifying this notion of easy enough to simplify it into what the statement holds under particular notion. Namely, as we find out, the above double-descent replicating formula can actually be realized by the parameter-wise error, comparing the parameter between the least-square solution and the actual concept. The formula for this is expressed by the exposed MSE on parameter function, for $\beta_{p,k}$ the coefficient value estimated for the k -th selected feature when the model is restricted for randomization of p features,

$$\text{MSE}_{\text{param}} = \|\hat{w} - w\|_2^2 = \sum_{j=1}^d (\hat{w}_j - w_j)^2 \quad (22)$$

where $\hat{w}_j$ is chosen such that it is equal 0 for $j \notin S$, otherwise $\beta_{p,k}$ for $S = \{i_1, \dots, i_p\}$ is the set of selected feature indices. Nevertheless, it is then found dubious why this is true, as for the original statement of Lafon and Thomas 2024 indicates correlation to the prediction error of the result, $x^\top \hat{w}$ instead. The result of evaluating $\text{MSE}_{\text{param}}$ is presented in Figure 8.

If, however, we turn our attention to the statement itself, of which refers to the prediction error MSE_{pred} , defined by the equation

$$\text{MSE}_{\text{pred}} = \frac{1}{n} \|\gamma - X_p \hat{\beta}\|_2^2 = \frac{1}{n} \sum_{i=1}^n \left(\gamma^{(i)} - x_p^{(i)T} \hat{\beta} \right)^2 \quad (23)$$

on h and c accordingly, a strong case of this happens to be not following the pattern indicted by the theorem. Rather than being the interpolation point of return, $p = n$ is now the convergence point of the prediction error instead. Test result is shown in Figure 9.

Thence, the theorem gives us two things – the interpolation point, in this setting, refers to the parameter error for their double descent behaviour. However, on the prediction results'

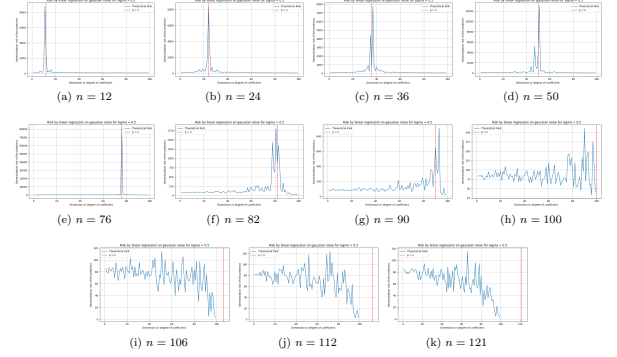


Figure 8. Contrary to Theorem 7.1, this is the behaviours on randomized setting for standard parameter-wise mean square error measure (MSE-on-parameter). For this model, we have $d = 100$, $\sigma = 0.5$, and variational n . Full test cases are for $p = [1, 100]$, $n = \{12, 24, 36, 50, 76, 82, 90, 100, 106, 112, 121\}$ accordingly. The test function of the concept itself is the same linear model, but with different input only.

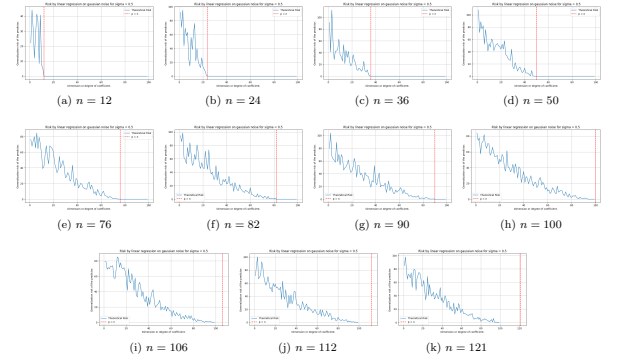


Figure 9. Contrary to Theorem 7.1 and both figure 8, this is the behaviours on randomized setting for standard parameter-wise mean square error measure (MSE-on-prediction). Especially, we do not see double descent pattern in this setting, as it is. For this model, we have $d = 100$, $\sigma = 0.5$, and variational n . Full test cases are for $p = [1, 100]$, $n = \{12, 24, 36, 50, 76, 82, 90, 100, 106, 112, 121\}$ accordingly. The test function of the concept itself is the same linear model, but with different input only.

error measure, the interpolation point correlates to the point where the defined generalization risk in such case equals zero. These patterns hold up to certain point. However, the test case is only up to $d = p$. An unexplored test case for $p > d$ indicate the surprising fact according still, to the theorem: the generalization risk takes double descent as a region with disturbance in the middle, but bias-variance appears afterward. This is illustrated in Figure 10, where the same theorem is applied but in the domain which is usually called as *overparameterized region*.

Such can then be seen as a prediction that double descent in certain case, can be extrapolated to be the transition between the two curvature of bias variance, or the abnormal point on the error curve measure. Such must then be verified, both in the experiment's correctness, and so on that lead to such result. Nevertheless, if true, it would mean that double descent indeed has another implication of itself.

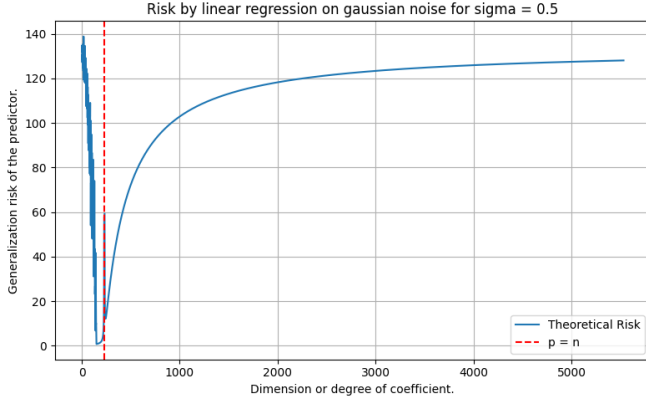


Figure 10. Similar experiment according to setting and result of Theorem 20, where p is in the range $[1, 5523]$. Here, we can see small double descent modelled exactly at the interpolation threshold, while the later region exhibit similar phenomenon as bias-variance tradeoff.

7.2.1 Extension

The current theory prompted for a more general treatment of the linear function class, as in-line with exploring reasonable existing alternatives in interpretation (Nakkiran 2019b).

Let us states some hypothesis about the previous experimental cases. First, is the lack of the begin-tail of the prediction on double descent of linear random cutoff features model. This can be explained informally as to be resulted from the uniformity of the operating space. Specifically, all parameters of the $\mathbb{R}^n$ parameter space, with respect to the dual space on such $f: \mathbb{R}^n \rightarrow \mathbb{R}$ projection is defined, are of the same order of magnitude. That is, for $f_{i,k}(x), f_{j,k}(x)$ of arbitrary parameter indexing on the set k of shared parameter choice, and $i \neq j$ being the distinct choice, then we say $f_{i,k}(x) = \mathcal{O}(f_{j,k}(x))$ of the growth of both function on value ranges. Approximately speaking, the choice of parameter does not dramatically growth to a point of no return, like in later examples of polynomial approximation model. Such special model of multi-variable linear projection of scalar output also reinforces the numeric growth of the numerical encoding expression on the hypothesis model. The behaviour of the MSE-on-parameter, the internal risk, is also explainable using the same notion, such is that for a black-box case approximation, such linear model does not allow for average parameter deviation per approximation to be over-saturated and skewed (for example, the degree of polynomial directly counteract this, by separating expressive complexity by growth degree). How such hypothesis can be evaluated, we will have to make it for the general extension experiment, which we will have to conduct.

Let us define the more general class of linear model, denoted $\mathcal{H}_L$, in the following structure of the linear model. We also note that the previous linear class $\mathcal{H}_p$ only solves for the problem of underparameterized, hidden parameters p' . Let $k \in \mathbb{N}$ denote the number of input vectors, and let $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ represent the input vectors where each $\mathbf{u}_i \in \mathbb{R}^q$ for $i \in \{1, 2, \dots, k\}$. Here, $q \in \mathbb{N}$ denotes the dimensionality of each input vector, with the convention that $q = 1$ corresponds to

scalar inputs (i.e., $u_i \in \mathbb{R}$).

For a given dimension parameter $d \in \mathbb{N}$, we define $\mathcal{H}_L$ as the set of all functions $h: (\mathbb{R}^q)^k \rightarrow \mathbb{R}$ that can be expressed in the unified form:

$$h(\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k) = \sum_{j=1}^d w_j \cdot \phi_j(\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k) + b \quad (24)$$

where $\mathbf{w} = [w_1, w_2, \dots, w_d]^\top \in \mathbb{R}^d$ is the weight vector, $b \in \mathbb{R}$ is the bias term, $\phi_j: (\mathbb{R}^q)^k \rightarrow \mathbb{R}$ is a feature map for $j \in \{1, 2, \dots, d\}$ that transforms the input vectors into a scalar feature, we consider the following important special cases, distinguished by their choice of feature map ϕ_j . If we do not have such feature mapping of which preserves relative linearity, then we said it is the basic linear class dynamic.

Definition 7.2 (Linear class, $q = 1$). Consider class $h: (\mathbb{R}^{d+1})^q \rightarrow \mathbb{R} \in \mathcal{H}_{\text{Lin}}$ of all functionals of the form as Equation 24. Then, when $q = 1$ of scalar observables, i.e., $u_i \in \mathbb{R}$ for $i \in \{1, \dots, k\}$, then the function is simply as:

$$h(u_1, u_2, \dots, u_k) = \sum_{j=1}^k w_j \cdot u_j + b \quad (25)$$

for $\phi_j(u_1, \dots, u_k) = u_j$ for $j \in \{1, 2, \dots, k\}$, i.e. deactive feature map.

Such class can be extended rigorously, through a varied change in dynamics, specifically, considering the general feature representation on ϕ .

Definition 7.3 (General feature, kernel linear model). Consider class $\mathcal{H}_{\text{Lin}}$. For maximum flexibility, allow arbitrary feature maps $\phi: (\mathbb{R}^{d+1})^q \times \mathbb{R}^k$ for $k = \{1, 2, \dots\}$ as of hidden or feature mapping processor internally. The functional form is then expressed by

$$h(\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k) = \sum_{j=1}^d w_j \cdot \phi_j(\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k) + b \quad (26)$$

for $j \leq d$ as the iterative form.

Common choices for such kernel mapping includes:

- *Polynomial features:* $\phi_j(\mathbf{u}) = u_{i_1} u_{i_2} \dots u_{i_m}$ for various index combinations
- *Radial basis functions:* $\phi_j(\mathbf{u}_1, \mathbf{u}_2) = \exp(-\gamma \|\mathbf{u}_1 - \mathbf{u}_2\|^2)$
- *Fourier features:* $\phi_j(\mathbf{u}) = \sin(\omega_j^\top \mathbf{u})$ or $\cos(\omega_j^\top \mathbf{u})$
- *Neural network activations:* $\phi_j(\mathbf{u}) = \sigma(\mathbf{v}_j^\top \mathbf{u} + c_j)$ for activation σ ,

thus basically hints at the generality of, at least being a *linear approximation* on various hypothesis spaces, while not actually being similar. Then, our setting of sparsity being synthetically induced introduces the basin of subclasses. For any $p \in \{1, 2, \dots, d\}$, we can then simply define the sparse subclass $\mathcal{H}_{L,p} \subseteq \mathcal{H}_L$ as the set of all functions $h \in \mathcal{H}_L$ where the weight

vector $\mathbf{w} \in \mathbb{R}^d$ has exactly p non-zero entries. Formally, per Lafon and Thomas 2024,

$$\mathcal{H}_{L,p} = \{h \in \mathcal{H}_L : |\{j : w_j \neq 0\}| = p\} \quad (27)$$

The subset $S \subseteq \{1, 2, \dots, d\}$ of indices with $|S| = p$ corresponding to non-zero weights is assumed to be fixed throughout the learning process (i.e., the sparsity pattern is predetermined). This constraint yields:

$$h(\mathbf{u}_1, \dots, \mathbf{u}_k) = \sum_{j \in S} w_j \cdot \phi_j(\mathbf{u}_1, \dots, \mathbf{u}_k) + b \quad (28)$$

where only p features contribute to the final prediction. This effectively masks the data randomly, simulating the general observation-based information corruptions, without inducing specific bias like *dropout*, though potentially leads to strong perturbation in cases of meaningful information threshold for different function classes.

8. Reformation of statistical setting

Most of the paper will argue about different features of statistical learning theory (Sterkenburg 2024; Vapnik 1999; Hajek and Raginsky 2021). It is then imperative to discuss the background setting of such. We are given the observations, or dataset of the form $\mathcal{S} = (\mathcal{X}, \mathcal{Y}) \subset \mathbb{R}^n \times \mathbb{R}^m$, of all 2-tuple pairs, assumed to be sampled or observed and governed by a distribution $\mathcal{D}$.

$$\mathcal{S} = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\} \subset \mathcal{X} \times \mathcal{Y}$$

The dataset is assumed to be i.i.d. sampling according to $\mathcal{D}$, which is unknown by the priori. Hence, we have the first set of assumption.

Assumption 8.1. *The observational space is governed by a particular sampling distribution $\mathcal{D}$, for fixed y -response according to the concept $c \in \mathcal{C}$. The sampling process is further assumed to hold no further structure, hence is identically and independently distributed (i.i.d.).*

The learning problem is then formulated as followed. Given the machine learning model expressed a hypothesis h of the hypothesis class $\mathcal{H}$, the learning theory aims for creating a procedure to learn either elements of the concept class $\mathcal{C}$ of all concepts $c : \mathcal{X} \rightarrow \mathcal{Y}$, or the function class $\mathcal{F}$ of all functions $f : \mathcal{X} \rightarrow \mathcal{Y}$, which is usually set to $\{0, 1\}$ or $[0, 1]$. This distinction is trivial, hence, if it is clear, we will talk about the concept class $\mathcal{C}$ only as representative form. The learner $\mathcal{L}(h)$ consider the set of possible hypothesis $\mathcal{H}$, in which might not coincide with $\mathcal{C}$. It receives a partial image of sample $\mathcal{S} = (x_1, \dots, x_2, \dots, x_n)$ drawn i.i.d. according to $\mathcal{D}$ as well as the label $(c(x_1), \dots, c(x_n))$. This constitutes the dataset $\mathcal{S}$, which are based on specific concept $c \in \mathcal{C}$ for the hypothesis to learn. The task is then to use (or extract) meaningful information to select a hypothesis $h_{\mathcal{S}} \in \mathcal{H}$ that accurately mimic c , with marginal error Θ . The notation $h_{\mathcal{S}}$ stands for all hypothesis that can be inferred from the range of the dataset (the first argument, $\mathcal{X}$).

The marginal error Θ is considered of two parameters, the empirical error $\hat{R}(h)$ and the generalization error $R(h)$. We give the following definition of them.

Definition 8.1 (Empirical risk). *Given a hypothesis $h \in \mathcal{H}$, a target concept $c \in \mathcal{C}$, and a sample $\mathcal{S} = (x_1, \dots, x_m)$. For some particular $\epsilon > 0$, the **empirical error** or empirical risk of h is defined by*

$$\hat{R}_{\mathcal{S}}(h) = \mathbb{P}_{x \in \mathcal{S} \sim \mathcal{D}} [\ell\{h(x), c(x)\} \geq \epsilon] = \frac{1}{m} \sum_{i=1}^m \ell\{h(x_i), c(x_i)\} \quad (29)$$

The generalization risk is then defined as the extension of the empirical risk, of which instead of being evaluated on the set $\mathcal{S}$, it is evaluated on arbitrary infinite density sample space. The notation for the risk calculation can also be $R_{\mathcal{S}}$ instead, of which the need is to classify the set of which it is evaluated from.

Definition 8.2 (Generalization risk). *Given a hypothesis $h \in \mathcal{H}$, a target concept $c \in \mathcal{C}$, and an underlying distribution $\mathcal{D}$ on $\mathcal{X}$. For some particular $\epsilon > 0$, the generalization error or risk of h is defined by*

$$\begin{aligned} R(h) &= \mathbb{P}_{x \sim \mathcal{D}} [\ell\{h(x), c(x)\} \geq \epsilon] \\ &= \mathbb{E}_{x \sim \mathcal{D}} [\ell\{h(x), c(x)\}] \\ &= \int_{x \in \mathcal{D}} \ell\{h(x), c(x)\} dP(x) \end{aligned} \quad (30)$$

In essence, the empirical risk measure the hypothesis to observation error, and the generalization risk captures the difference of the hypothesis to the actual concept. Notice that by doing this, we have implicitly assumed that the observations and the concept is not the same thing, under the majority of scenarios. For the observations to be *approximately equal* or sufficiently captures the concept, there then would have to be various criteria to fulfil. This can be illustrated as Illustration 11. For fixed $\mathcal{H}$, for fixed and sufficiently large $\mathcal{S}$, and no observation (data) errors, the empirical risk is the generalization risk. These two measures between h and c constitute the learning problem, which can also be separated into both cases – either empirical learning or generalization learning, one to minimize $\hat{R}(h)$, and one to minimize $R(h)$. Here, we emphasize such hierarchy by defining the generalization goal as the **nested criteria** for empirical risks minimization.

Definition 8.3 (Empirical learning problem). *We present the formal form of the empirical learning. Suppose we have a **target**, $c \in \mathcal{C}$, where $\mathcal{C}$ is an arbitrary concept class that captures targets of the same type. Suppose we are provided a set of observations $\mathcal{S}$. The problem is to use certain algorithm $\mathcal{A}$ using $\mathcal{S}$, to obtain a hypothesis h^* for a fixed $\mathcal{H}$ such that:*

$$R(h^*) = \min_{h \in \mathcal{H}} \hat{R}(h) = \min_{h \in \mathcal{H}} \mathbb{E}_{x \sim \mathcal{D}, x \in \mathcal{S}} \ell\{h(x), c(x)\} \quad (31)$$

The generalization best h is also called in literature as the *Bayes model*, such that it reduces the Bayes risk that is the infimum of the generalization risk, $R^* = \inf_{h \in \mathcal{H}} R(h)$. The hypothesis h^* is often called the *empirical best*, for it being the minimal, finite hypothesis of the lowest loss evaluation on the entire observation space $\mathcal{S}$. There exists no certified assumption regarding whether h^* aligns with the minimal generalization error.

Definition 8.4 (Generalization learning problem). *We present the formal form of the generalization learning problem. Suppose we have a **target**, $c \in \mathcal{C}$, where $\mathcal{C}$ is an arbitrary concept class that captures targets of the same type. Suppose we are provided a set of observations $\mathcal{S}$. Supposed we have an algorithm $\mathcal{A}$ that for fixed hypothesis space $\mathcal{H}$, equation 31 holds true. The problem is to use certain algorithm $\mathcal{A}'$ such that, under limited availability, to obtain h , satisfies:*

$$R(h) = \min_{h \in \mathcal{H}} R(h) \leq \{\epsilon\}, \quad \epsilon > 0 \quad (32)$$

For a set of risk bounds ϵ . If the setting is deterministic, then there exists $\epsilon = 0$ accordingly.

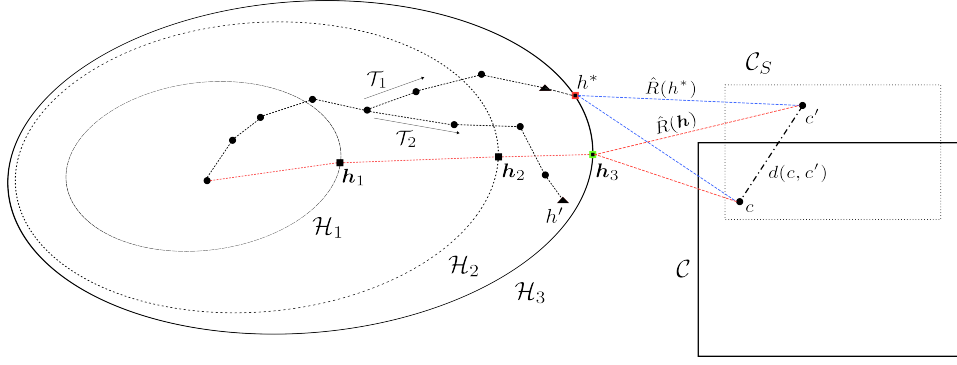


Figure 11. Illustrative dynamic of the learning problem. For an incremental hypothesis space sequence $\mathcal{H}_1, \mathcal{H}_2, \mathcal{H}_3$, we aim to obtain the procedure $\mathcal{T}$ that would either reach the *generalization solution* h , or the *empirical solution* h^* for a particular hypothesis class, with respect to the concept c and its observed concept c' . Model selection hence dictates how an algorithm or procedure than choose the best possible hypothesis to approximate the generalization solution from the empirical solution set. Do notice that here we explicitly state that the data would create a **proxy concept** that overlaps with the concept class and of some arbitrary ‘distance’ from the true concept by $d(c, c')$. In case the two classes overlap, there still exists the arbitrary distance. Also, we can also observe the intuitive notion of increasing hypothesis class - while it indeed can help in getting closer to the concept class, the somewhat intrinsic property of current learning theory being the relative random procedure class make it probabilistically unstable.

Then, we can partially say that generalization problem is an advancement from empirical solution. As such, empirical solution imposes the observed concept, and not the concept itself. This is reflected in modern machine learning landscape, by modelling the concept as distributions, and then using notions such as KL-divergence to measure such relative disparity.

The process of generalization goal can be, described in certain form as the global optimization objective in which empiricism can obtain from reasonable set of assumptions and conditions. The following algorithmic framing indeed aims to focus and extradite such sense of operational goal.

Optimization problem 2: Empirical-generalization learning dynamics (non-proxy)

Data: $(x_i, y_i) \in \mathcal{S}, \Gamma \supset \mathcal{S}$, objective $\ell(y, f(x)), \nabla(\cdot)$ differentiable gradient field on ℓ .

Result: Expected optimized model to the minimal error on the objective $\ell(\cdot, \cdot)$, dependent on $\mathcal{S}$.

begin

$\Gamma \supset \mathcal{S}$, the general set;

$R(\cdot), \hat{R}(\cdot) \leftarrow \ell$ as objective functional;

for Fixed $\mathcal{H}$, with $h \in \mathcal{H}$, for $\epsilon > 0$ **do**

Find $h \rightarrow h'$, using algorithm $\mathcal{A}$ for minimization objective, such that

$$\mathcal{A} \left(\min_{h \in \mathcal{H}} \hat{R}_{\mathcal{S}}(h) + R(h) \right) \leq \epsilon \quad (33)$$

for procedure $\mathcal{A}(\cdot, \cdot)$ of structural constraints;

end

end

return Optimized model h on arbitrary objective space (ℓ, ∇) .

These two stages of learning procedure comes of as the intrinsic design of machine learning setting. As opposed to interpo-

lation, where empirical learning problem would have to be put into first priority to achieve the best possible point-through model, machine learning leans on approximation more and a separated criterion that is only visible in the learning setting - the fact that we are using c' to extract c . This disparity is exactly the **generalization problem**, of which is then also the goal of machine learning, to approximate the general solution using the observed solution. One then can ask what is the difference between the two notions, and when we can guarantee that both will ‘converge’ to the same point. The process of **model selection** is then the procedure that will optimize $h \in \mathcal{H}$ to a given target fixture, such as the empirical best.

One of the major assumption or observation from the point of the hypothesis class, is that the hypothesis cannot know the generalization error. Thereby, the more efficient and often used procedure/algorithm in training, is then the *empirical risk minimization* (ERM) method, minimizing only the empirical risk to the empirical best; then, certain measures to ‘generalize’ this heuristically is apprehended on the hypothesis. The process of model selection is also the place that we get the concept of **bias-variance tradeoff**, and its longstanding form to be a foundational metric on model selection.

8.1 Empirical and structural algorithms

Suppose that for a concept class $\mathcal{H}$, we can then partition it to $\mathcal{H} = \bigcup_{k \in \mathbb{N}} \mathcal{H}_k$ for finite k hypotheses. Assume that they satisfy the uniform convergence (Shalev-Shwartz et al. 2010), which means in the arbitrary path landscape, one can guarantee ℓ -reduction, of m samples on $\mathcal{S}$ to:

$$\lim_{m \rightarrow \infty} \sup_{\mathcal{D}} \mathbb{E}_{\mathcal{S} \sim \mathcal{D}^m} \left[\sup_{h \in \mathcal{H}} [R(h) - R_{\mathcal{S}}(h)] \right] = 0 \quad (34)$$

Then such is said to be possible as a ruling dynamic when for a learning problem, the empirical risks of hypotheses in the

hypothesis class converges to their population risk uniformly, with a distribution-independent rate. The result of this is that a problem then can be considered learnable with the ERM rule, for any given characterization. We state the following definition.^d

Definition 8.5. *A learning problem is learnable if there exist a learning rule $\mathcal{A}$ and a monotonically decreasing sequence $\epsilon_{\text{cons}}(m)$ such that $\epsilon_{\text{cons}}(m) \xrightarrow{m \rightarrow \infty} 0$ and*

$$\forall \mathcal{D}, \quad \mathbb{E}_{S \sim \mathcal{D}^m} [F(\mathcal{A}(S)) - F^*] \leq \epsilon_{\text{cons}}(m), \quad (35)$$

with $F^* = \inf_{h \in \mathcal{H}} R(h)$. A learning rule $\mathcal{A}$ for which this hold is denoted as a *universally consistent learning rule*.

There are certainly a fundamental class of rule or algorithm $\mathcal{A}$ to resolve this particular uniform convergence of classes. One of the major assumption or observation from the point of the hypothesis class, is that the hypothesis cannot know the generalization error. Thereby, one strategy is the *empirical risk minimization* (ERM) method, minimizing only the empirical risk to the empirical best; then, certain measures to ‘generalize’ this heuristically is apprehended on the hypothesis.

Definition 8.6. *A rule $\mathcal{A}$ is an ERM (Empirical Risk Minimizer), denoted $\mathcal{A}_{\text{ERM}}$ if it minimizes the empirical risk $\hat{R}(h) = (1/m) \sum_{c \in \mathcal{C}} \ell(h, c)$, over the observational space S ,*

$$\hat{R}_S(\mathcal{A}_{\text{ERM}}(S)) = \hat{R}_S(h_S) = \inf_{h \in \mathcal{H}} \hat{R}_S(h) \quad (36)$$

Hence, the ERM tries to obtain $\text{ERM}(h') = \arg \min_{h \in \mathcal{H}} \hat{R}(h)$.

The problem of overfitting still somewhat resides in the setting of the method. Hence, usually, we have an implicit term in addition, called the *regularizer*, of which is as followed.

Definition 8.7. *A rule $\mathcal{A}$ is an SRM (Structural Risk Minimizer), denoted $\mathcal{A}_{\text{SRM}}$ if it minimizes the empirical risk $\hat{R}(h) = (1/m) \sum_{c \in \mathcal{C}} \ell(h, c)$, over the observational space S ,*

$$\hat{R}_S(\mathcal{A}_{\text{SRM}}(S)) = \hat{R}_S(h_S) + \lambda r(h) = \inf_{h \in \mathcal{H}} \hat{R}_S(h) + \lambda r(h) \quad (37)$$

of the regularizing term $r(\cdot)$. Hence, the SRM tries to obtain $\text{ERM}(h') = \arg \min_{h \in \mathcal{H}} \hat{R}(h)$, while taking into consideration a forcing term r , usually defined on the model complexity.

Hence, the SRM is inherently stronger than ERM. The regularizer $\lambda r(h)$, in specific case of convex loss landscape and reasonably well-behaved irregularities between domains, and of the structural regularization term $r_s(h) = \sum_{i,j} w_{ij}^2$ for weights of ij index, then we have that

$$\text{ERM} \left(\inf_{h \in \mathcal{H}} \hat{R}_S(h), \ell \right) \approx o \left(\text{SRM} \left(\inf_{h \in \mathcal{H}} \hat{R}_S(h) + \lambda r(h), \ell + \lambda r(h) \right) \right) \quad (38)$$

d. The reason why $r(\cdot)$ is not characterized in this preliminary section, is because of their uncertainty of effect. Hypothetically, $r(\cdot)$ is an additional structure (bias) of which apply a new gradient field in conjunction to the loss gradient field itself. This addition is uncertain of its content and analysis, since it depends on various stochastic dynamics of the model. We reserve from using regularization for now because of such reason.

Where the little- o notation is used for saying of the optimization strength per arbitrary time-similar notion of algorithmic evolution. The reason for this change is because of the restriction domain that SRM indicted on the model optimization landscape. Effectively, we introduce the structural bias $\lambda r(h)$ that forces the morph of the total loss landscape. In well-behaved case, this is usually effective to a point, however in realistic system and pathological cases, such can lead to the c' approximation in c -space, just as we introduced in the reformulation of learning system. In this paper, we would like to presume the ERM method in question as the famous *gradient descent*. We refer to Achlioptas, n.d.; Ruder 2017 for general view of the algorithm, and J. Zhang 2019 for a source on particularly gradient descent algorithm on deep learning models.

8.2 Overfitting and underfitting

Following the above framework of statistical probabilistic learning theory, one understanding about the concept of underfitting and overfitting (similar to James et al. 2009) can be reached, using the notion of *partitioning* of observation space. Such work is typically done because of **practical condition**, in which the following constraints is applied.

Assumption 8.2. *No knowledge of $c \in \mathcal{C}$ is provided.*^e

Without such knowledge, $R(h)$ cannot be computed, accordingly. We specify such approach as *heuristic*, as it is only an approximation, and for better definition, we define it as followed.

Definition 8.8 (Heuristic empirical risk). *Given a dataset S of size m , supposedly of the concept $c \in \mathcal{C}$, and the hypothesis $h \in \mathcal{H}$, then the **heuristic empirical risk** take a partitioning $k < m$ with $k/m \geq 1/2$, hence defining the heuristic empirical loss on k , denoted $\hat{R}_k(h)$.*

Definition 8.9 (Heuristical generalization risk). *Given a dataset S of size m , supposedly of the concept $c \in \mathcal{C}$, and the hypothesis $h \in \mathcal{H}$. Fixed $\hat{R}_k(h)$, for the final optimization of algorithm $\mathcal{A}$, then the **heuristic generalization risk** take the remaining 2-partition of portion $p = k - m$, hence defining the heuristic generalization loss on p , denoted $\hat{R}_p(h_{\mathcal{A}})$.*

The two heuristic definition however, can be noted to not be equal to the true generalization risk – empirical risk pair in reality. Nevertheless, one can then define overfitting and underfitting in such manner.

Definition 8.10 (Underfitting). *A model h is called **underfitting** if for a constant $\epsilon > 0$ chosen for the setting, then $\hat{R}_k(h_{\mathcal{A}}) > \epsilon$,*

e. Few points need to be said about this assumption. This assumption represents the *worst-case possible* of a system dynamic. From the perspective of a designer, inherent leakage of information can be provided in the design of a model. In such, case we are operating in the mode of a *resource-constrained, closed game*, in which the model with further assumptions of fixed class, operates on. The problem space itself is inherently large and penalize any additional structural changes. If one is to occur, we usually mention it as *implicit bias*, but another name that would be more fitting would be *structural attractor*, after the notion of gradient attractor in chaotic systems.

$\hat{R}_p(h_\alpha) > \epsilon$ over the entire observation space.

Usually, underfitting is defined to be such that test error is smaller than training error partition, however, we do not think that is a good definition. Especially since, at least in this setting, the empirical error under algorithm $\mathcal{A}$ is reduced to the representative smallest error that $\mathcal{A}$ can achieve on that particular model, within particular setting of h , then come the heuristic generalization error. Considering also that $k/m \geq 1/2$, that is, at least the training partition is half of the observation set, then only under very specific case will $\hat{R}_p(h_\alpha)$ be smaller than $\hat{R}_k(h)$, statistically speaking. Thence this line of thought also reinforces the idea that both of them will be larger than the threshold ϵ . Usually, heuristically, this threshold of error is 0.3, normalized. Next, we define the notion of overfitting.

Definition 8.11 (Overfitting). *A model h is called **overfitting** if there exists a specific constant $\epsilon > 0$ chosen for the setting, such that $\hat{R}_k(h) < \epsilon$, $\hat{R}_p(h_\alpha) > \epsilon$ over the entire observation space.*

Namely, this mean that the static learning makes the generalization error worse. This draws the analogy of the student - studying explicitly for the past paper tests, yet unaware of the larger margin of questions in the large set of the test paper pool, leading to subpar performance. Similarly, overfitting is theorized to be fitting too fit to the reduced observation space - hence when it comes to the generalization observation space, it fails within such heuristic. The notion works on the mask of the risks, $\hat{R}_k$ and $\hat{R}_p(h_\alpha)$.

In practice, Assumption 8.2 means we cannot have $c \in \mathcal{C}$, or its probabilistic interpretation $\mathcal{D}$ such that $x \sim \mathcal{D}$, or in the extreme non-structured case, $(x, y) \sim \mathcal{D}$. In such case, the typical practical fix relies on the partitioning of the available partitioning space, that is, the empirical-generalization proxy. For large enough singleton count of $|\{(X_k, Y_k)\}|$, we assume that $\hat{R}_{\text{Gen}}(h) \approx R(h)$. Thus, the algorithm can be stated in a form as Algorithm 3.

The assumption above of the generalization set to the generalization truth error is apparent in the usual system theorem, Theorem 8.3.

Theorem 8.3 (Mohri, Rostamizadeh, and Talwalkar 2012). *For fixed $\mathcal{H}$, for fixed and sufficiently large $\mathcal{S}$, and no observation errors, the empirical risk is the generalization risk:*

$$\mathbb{E}_{\mathcal{S} \sim \mathcal{D}^m} [\hat{R}_{\mathcal{S}}(h)] = R(h) \quad (40)$$

Proof. By linearity of the expectation, and the fact that we assume the sample is given i.i.d., we can write:

$$\begin{aligned} \mathbb{E}_{\mathcal{S} \sim \mathcal{D}^m} [\hat{R}_{\mathcal{S}}(h)] &= \frac{1}{m} \sum_{i=1}^m \mathbb{E}_{\mathcal{S} \sim \mathcal{D}^m} [\mathbb{1}_{h(x_i) \neq c(x_i)}] \\ &= \frac{1}{m} \sum_{i=1}^m \mathbb{E}_{\mathcal{S} \sim \mathcal{D}^m} [\mathbb{1}_{h(x) \neq c(x)}], \end{aligned} \quad (41)$$

for any x in sample $\mathcal{S}$. Thus, we result in

Optimization problem 3: Empirical-generalization learning dynamics (proxy)

Data: $(x_i, y_i) \in \mathcal{S}$, $y_i = \{\pm 1\}$, hinge objective $\ell(y, f(x))$, $\nabla(\cdot)$ differentiable gradient field on ℓ .

Result: Expected optimized model to the minimal error on the objective $\ell(\cdot, \cdot)$, dependent on $\mathcal{S}$.

begin

$((X_1, Y_1), \dots)_n \leftarrow \text{Partition}_n(\mathcal{S});$

$\{(X_j, Y_j)\} \leftarrow \text{Train};$

$\{(X_k, Y_k)\} \leftarrow \text{Gen}$ for $k \neq j$;

for Fixed $\mathcal{H}$, with $h \in \mathcal{H}$, for $\epsilon > 0$ **do**

(Optimistically) achieve $h \rightarrow h'$, using algorithm $\mathcal{A}$ for minimization objective, such that

$$\mathcal{A} \left(\min_{h \in \mathcal{H}} \hat{R}_{\text{Train}}(h) + \hat{R}_{\text{Gen}}(h) \right) \leq \epsilon \quad (39)$$

for procedure $\mathcal{A}(\cdot, \cdot)$ of structural constraints.;

end

end

return Optimized model $h_{\mathcal{A}}$ on arbitrary objective space (ℓ, ∇) .

$$\begin{aligned} \mathbb{E}_{\mathcal{S} \sim \mathcal{D}^m} [\hat{R}_{\mathcal{S}}(h)] &= \mathbb{E}_{\mathcal{S} \sim \mathcal{D}^m} [\mathbb{1}_{h(x) \neq c(x)}] \\ &= \mathbb{E}_{x \sim \mathcal{D}} [\mathbb{1}_{h(x) \neq c(x)}] \\ &= R(h). \end{aligned} \quad (42)$$

which yields the desired result. $\square$

This theorem can be transferred to a more generalized conjecture, of which fit the generalized notion of the empirical proxy and so on.

Conjecture 8.1. *Supposed $\mathcal{H}, \mathcal{C}$ are fixed, for $\mathcal{S}$ is the empirical c^* for $c \in \mathcal{C}$ the true concept. Then, for algorithm $\mathcal{A}(\cdot, \cdot)$ enforcing minimization apparatus, and the specific objective on the second argument, we have:*

$$\begin{aligned} \mathcal{A}_{j \rightarrow \infty, \mathcal{S}_i \neq \mathcal{S}_k} \left(\min_{h \in \mathcal{H}} \hat{R}_{\text{Train}}(h) + \hat{R}_{\text{Gen}}(h), [\ell] \right) \\ \approx \mathcal{A} \left(\min_{h \in \mathcal{H}} \hat{R}_{\mathcal{S}}(h) + R(h), [\ell] \right) + \epsilon_{\mathcal{S}} \end{aligned} \quad (43)$$

For $\mathcal{S}_i \neq \mathcal{S}_j$ such that there are only distinct samples, j the observation set size, $[\ell]$ the set of structural or empirical objectives enforced on the problem landscape, and $\epsilon_{\mathcal{S}}$ is the empirical minimum that the algorithm result can achieve assuming sufficient 'perfect interpolation'.

By the end of this section, we would also want to note that we deliberately choose an expression basis of $\mathcal{A}$ via the couple tuple $(\min, [\ell])$ as a choice for algorithm-agnostic class of algorithm satisfies the dynamic only, not just implementations. The

algorithm, in certain sense, depends much more on structural assumptions, of which is not accessible via observations but via design, and intrinsic information specified in the observation class itself. This is a requirement for fine-tuning analysis in the long-run, as there exists a myriad of different approaches.

9. Complexity notion

Assume the working space is the real field $\mathbb{R}^n$. Let us recall a model $h \in \mathcal{H}$ of certain hypothesis class $\mathcal{H}$. Then, the **complexity** refers to the complexity of the hypothesis h itself; there is also the notion of class maximal complexity for $\mathcal{H}$. Complexity in this case refers to the mass of the model, or rather, to answer the following questions:

1. What is the effective expression range of a hypothesis?
2. What is the hypothesis structured in?
3. If the model is constructed by parameters, how many parameters are there?
4. How many components are there in specific model h , and how would another model $h' \in \mathcal{H}$ differs from h ?

Those questions are not so easily answered. Specifically, analysis often puts the hypothesis class $\mathcal{H}$ as a class of functions. So, there exists the class $\mathcal{H}_n$ of n -th degree polynomials, or else. Effective complexity in such case means the range of expression a function can give – for example, for n -th polynomial class. For the class $\mathcal{H}_n$ of polynomials of degree n on a closed interval $[a, b] \subset \mathbb{R}$, the Weierstrass approximation theorem guarantees that, as $n \rightarrow \infty$, any continuous function $f : [a, b] \rightarrow \mathbb{R}$ can be approximated arbitrarily closely in the uniform norm. Hence, the expressive complexity of $\mathcal{H}_n$ grows with n , but for fixed n only a finite-dimensional subset of $C([a, b])$ is exactly representable. This distinction highlights the difference between *approximation capacity* of a hypothesis class and the *exact representational capacity* captured by its parameterization. These expressions depend on the field or space the hypothesis class lives in, for example, if the hypothesis is configured to work in the set of all functions in $\mathbb{F} = \mathbb{R}$ of algebraic field, then we call it the algebraic family. There are various hypothesis classes in such case, and usually, not a lot of them can be analysed and represented in the same way, and hence also the analysis of the ‘size’ of the hypothesis, also between different hypothesis classes, prompted the usage of a more general notion, called **representation complexity**^f.

The above complexity (and so is in Table 1), aside from sample space-related measure, can also be applied onto the concept

f. Representation complexity refers to the number of components needed to express h and generally, any hypothesis in $\mathcal{H}$. This is often the parameters count, since the parameters themselves are used in constructing the function in algebraic system. Thus, again, an n -th degree polynomial might have $2(n+1)$ parameters, for $n+1$ variational input (or variable), and $n+1$ coefficients. Notice that we specifically at the beginning, use the notion of algebraic structure $\mathbb{R}$ because of it. If we restrict ourselves to the expression, for example, for $\mathbb{B} = \{0, 1\}$, then it can be expressed in various ways in logic theory, such as conjunctive normal form (CNF), disjunctive normal form (DNF), and else. This forbid the analysis based on the effective process itself, but rather the number of components required, hence taking them as complexity of construction.

class’s members itself. We also take the class complexity in such case, as the argument that maximize those measure: that is, the largest value of the measure, taken on the class as the class’s complexity.^{ghijk}

The next complexity measures that can be taken of, comes from the learning process structure itself. Usually, in statistical learning theory (Hajek and Raginsky 2021; Mohri, Rostamizadeh, and Talwalkar 2012; Shalev-Shwartz and Ben-David 2014), we often use the setting that is related to **supervised process**. This usually means the process can be configured in a feedback loop, or induced response toward the model – often come from the observational space, where the concept that is set to be the target of the process – then called as learning process – is conjured. For now, we reserve this to the classical learning theory and process, and will not inquire further of such direction.

The learning process assumes an operational space containing the structure, for example, $\mathbb{R}$ -encoding space of real values, and a set of **samples** living in such space, denoted $\mathcal{S}$. Those samples are often referred to as the **observation set** of a particular objective of learning. As mentioned above, $\mathcal{S}$ belongs to the concept c , assumed to be of certain concept class $\mathcal{C}$. $\mathcal{S}$ is obtained by using an observing process to create or generate the result. This process can be determined or considered in multiple way, however, generally can be considered by having a deterministic interpretation (using an explicit function $c(x)$) or by probabilistic approach (modelling a distribution $\mathcal{D}$). Hence, we gain another type of complexity, with regard to the process called **sample complexity** – usually defined of cardinality – the amount of sample needed for particular purpose, for particular objective. For the complexity category up to the structure of the learning setting by itself, we call it the **sample space complexity**.

One final complexity that is worth mentioning is then the **optimization complexity**, measure on the action of the learning algorithm. By such, it investigates the construction of an optimization algorithm solely dependent on its size of the algorithm, and other non-process related measures. Optimization complexity can also be related or expressed, in situation of time complexity, which is typical in computing system.

One might ask why we need to determine and categorize complexity measures and the need to generalize and differentiate of each one. Indeed, while in usual use cases, such complexity notion can be considered sufficient in controlling and gauging the model itself. However, just as James et al. 2009 indicated in the section on deep learning double descent, such complexity

g. Expressive complexity measures the variety of input-output behaviours the model can implement.

h. Structural complexity is decomposable into component-wise measures (e.g., per-layer, per-module) and useful for architectural comparisons.

i. Sample complexity depends on hypothesis class, loss, and desired error/confidence. Further clarification must be made of a grounded definition for utilization.

j. Sample space complexity infers the statistical measure over the sample space; affects learnability and effective sample size.

k. We also note that the notion of optimization is usually hard to pin down analytically, often characterized empirically or by worst-case bounds.

Table 1. Descriptions of complexity measures applied to models and learning processes, from Hajek and Raginsky 2021; Sugiyama 2015; Mohri, Rostamizadeh, and Talwalkar 2012; Kearns and Vazirani 1994 and Vapnik 1999 with some custom notions.

Type	Notation	Purposes	Notes
Representation	$\mathcal{M}_r$	Complexity of encoding a model (size of representation, alphabet-dependent).	Model-level measure.
Expressive	$\mathcal{M}_e$	Complexity of the model's observable behaviour (operational expressivity of the mapping it implements).	Model-level measure.
Structural	$\mathcal{M}_s$	Complexity of the model's internal architecture (how components compose and interact).	Model-level measure.
Structural mass	$\mathcal{M}_\Sigma$	Aggregate representation size derived from representation complexity across the model.	Sum of $\mathcal{M}_r, \mathcal{M}_e, \mathcal{M}_s$
Sample	$\mathcal{M}_{p \in \mathcal{S}}, \mathcal{M}_p$	Number of samples required to reach a prescribed performance level (statistical sample-count).	Process-level measure.
Sample space	$\mathcal{M}_\mathcal{S}$	Intrinsic complexity of the data distribution / sample domain (richness, diversity, structured-ness).	Global process measure.
Optimization	$\mathcal{M}_e$	Complexity of the learning algorithm (convergence rate, computational/iteration cost, sensitivity) or worst-medium-best learning cost.	Process-level measure.

Notes: The notion of structural mass is convoluted. In certain setting, it is simply the parameter space $\theta \in \Theta$, but in some setting, there exists fundamental difference than identically weighted parameter space complexity, as seen in multi-basis polynomial models, not taking into account homogeneous deep learning networks. Optimization complexity coincides with PAC-learning bounds for polynomial runtime, and *sample complexity* sometimes can only be the dimension of the input/observation space.

might eventually, of some regions, inadequate to express and evaluate the complexity of the model in its operations any more. Such transition is unanswered, and while the founding of complexity notion is still fragmented, it is indeed difficult to not paying attention specifically to those notions themselves.

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Appendix A. Reevaluation of theorem 20

To verify the theorem itself, we test it with a standard setting without the theorem results. This is reflected in Figure 7. The result is somewhat consistent with the theorem and its latter parameter error and prediction error results, however, this time, it correlates exactly as the prediction error instead. Additionally, peculiarity occurs in the non-double descent region, with much less volatility than in the prediction of the theorem.

Algorithm 4: Linear regression with Gaussian white noise, full control.

Data: n, p, d, σ, r , fixed range $[a, b]$, $u \in [0, 1]$

Result: Threefold error measure – MSE on parameters p, d , MSE on prediction Y', Y , and theoretical risks measure by prescription.

begin

Randomizer($\cdot$) $\leftarrow$ PCG64($n \leq 100$);

$\sigma \leftarrow 0.5, d \leftarrow 150$;

$nl \leftarrow \{12, \dots, 280\}$;

$p \leftarrow 150$;

$\mathbf{p} \leftarrow \text{range}(1, p)$;

$[a, b] \leftarrow [-100, 100]$;

$w_c \sim \text{PCG64.Uniform}([1, d], 150)$;

$b_c \sim \text{PCG64.Uniform}([1, 10])$;

$w_p \sim \text{PCG64.Uniform}([1, p], p)$;

$b_p \sim \text{PCG64.Uniform}([1, 10])$;

$X \sim \text{PCG64.Uniform}(\text{range} \leftarrow [a, b], \text{size} \leftarrow [n, d])$;

$\text{concept}(\cdot, \cdot, \cdot) \leftarrow (d, w_c, b_c)$;

$Y \leftarrow \text{concept}(X, w_c, b_c)$;

$\sigma_{\text{scaled}} \leftarrow \sigma \cdot \mathbb{E}[|Y|]$;

$Y \leftarrow Y + \text{PCG64} \dots \text{Normal}(\mu \leftarrow 0, \sigma \leftarrow \sigma_{\text{scaled}}, \text{size} \leftarrow |Y|)$;

for $n \in nl$ **do**

$w_p^* \leftarrow \text{PCG64.Permutation}(d, p)$;

for $i \leftarrow 1$ **to** $|w_p|$ **do**

if $i \neq w_p^*$ **then**

$w_p[i] \leftarrow 0$, fixed zero point.

end

end

$\text{model}(\cdot, \cdot, \cdot) \leftarrow (d, w_p, b_p)$;

$Y \leftarrow \text{model}(X, w_p, b_p)$;

$\text{approx} \leftarrow \text{LeastSquare}(\text{model}, Y', X, Y)$, fixed w_p .

end

$\text{MSE}_{\text{param}}(\cdot, \cdot, \cdot) \leftarrow \text{MSE}(Y, \{\text{approx}\}, \text{model})$;

$\text{MSE}_{\text{pred}}(\cdot, \cdot) = \text{MSE}(Y, Y') \hat{R}_{\text{Thm}} = \text{TR}(w_c, \sigma, p, d, n)$;

end

return $\text{MSE}_{\text{param}}, \text{MSE}_{\text{pred}}, \hat{R}_{\text{Thm}}$

Randomizing pseudo-random engine

Common presetting for parameters

Common full-list n count of datapoints

Underparameterization to equal

n samples of d dimensions.

Noisy featuring

We use the Algorithm 4 for verification, including three modes of error measure, which will then be calculated accordingly. For gradient descent, we refer to another experiment. For such algorithm, we notice that the operational space of the model is $\mathbb{R}^n \rightarrow \mathbb{R}$. The masking of parameters then is not well-received, since the output space is far more skewed – single dimension resultant with no scaling difference because the model in question is supposed to be linear. Regarding such, it makes sense to realize that the model itself can still attain relatively stable error, because its feature compounds on the scaling parameters

To test the stability of the model, we run for a stable section of the permutations done on p . The result is outlined in 7. Interestingly, while the interpolation threshold theorized being $p = n$ does not hold, the pattern of double descent without head descent still fundamentally exists, albeit in a much more constrained region, as can see from $n = 12$ to $n = 150$. The pattern from then onward follows a reduced trend, without the chaotic behaviours seen. The result is shown in Figure 12.

Appendix 0.1 Remark

At least from this experiment, overall, results gathered using Theorem 7.1 and variations of it suggests inconsistent results. The supposed interpolation threshold has different interpretation and indication in different scenario, including both parameter-wise error and prediction-only error measure. Furthermore, verification also shows inconsistency with the interpolation threshold,

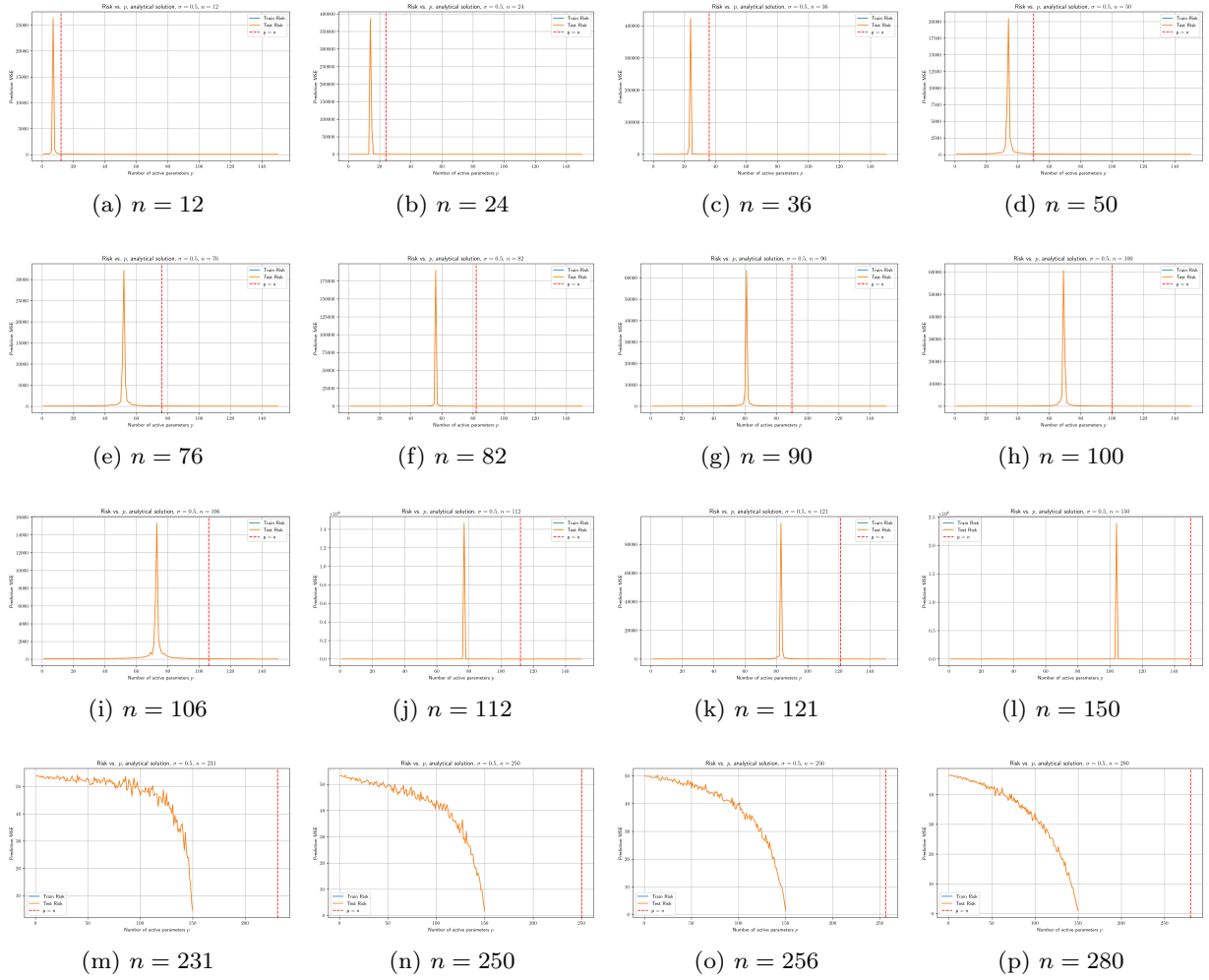


Figure 12. Risk curves for varying sample size n (dimension $d = 100$, noise $\sigma = 0.5$). Double descent is determined in similar sequence, even though the interpolation theoretical $p = n$ is actually shifted over to the right from the actual interpolation observables, in parameterization. Afterward, correlation fails, and the theoretical $p = n$ is ineffective, similar to previous theorem-based test run. Sample set size range is $n \in \{12, 24, 36, 50, 76, 82, 90, 100, 106, 112, 121, 150, 231, 250, 256, 280\}$. One additional remark is that the error landscape is relatively thin in either side, making it abnormal in comparison to actual bias-variance curvature.

albeit given the fact that the margin of error from the highest peak to the interpolation point is consistent up to certain n . Breakdown of the result from theorem, as well as double descent behaviour starts with high n counts, as well as interesting outlook of bias-variance reappearance for $p \sim 5000$, far more than what is tested in normal testing cases. Such result suggests further experiments to validate the observations, as well as statistical analysis of the entire system.

Specifically, the interpretation of the result is not perfect. Under such setting, then for the concept of the same class, so $d = 150$, any concept of $p > d$ would result in the metric taking zero value of the ‘truncated region’ where d is literally flat on the $d + 1$ onward space. Hence, the error being added on top of such error measure is the component distance of p to the flat sub $p - d$ space itself – which explains the curve going upward, as more and more risks are configured. Nevertheless, the amount of data provided, at least in terms of this model, does indeed have non-trivial effect on the model itself, and the least square solution of best fit which gives the potential minimal, hence the empirical best. Considering the results, we can see regions where the theorem ‘breaks’ of its bias-variance pattern. Further experiment to verify such claim and reconfiguration is needed. Worryingly, in this particular case, there exists no saddle point that was theorized by bias-variance tradeoff, which must then be explained accordingly. Furthermore, we also notice the troublesome fact that this particular pattern works on *same operational space* models, that is, linear models approximating linear models. This setting is fairly limited, and would not generalize well. However, we also do not know what kind of measure to take in such case. Conflicting notions and reports aid into the confusion of such topic more than can be normally observed, and the anomalous behaviours in some researches (for example, in Shi et al. 2024, we received that in GNN, there exists no trace of double descent) makes it even harder. As such, both the status of bias-variance being false,

albeit in a given range only, and double descent in masking certain behaviours of the modelling process, have taken a mysterious stance in the modern machine learning community.

While discovering the double descent phenomena, it also exposes theoretical concepts that are not organized or formalized, portions of the theory and empirical observations that are within conflict of each others, issues with definitions, theorems, and interpretation of them. Those problems would then ultimately hinder further research in such direction, as well as broader theoretical requirement of the theory of machine learning, and specifically to a larger and more complex system of deep learning.

Appendix B - Observing gradient descent decomposition

While the interpretation of what the bias-variance tradeoff, the usual training-versus-test curve, and statistical learning consideration, another interesting aspect to consider (as well for double descent), is how bias-variance decomposition acts on gradient descent and its components. This claim is perhaps reinforced by the work of Adlam and Pennington 2020 which indicates that different bias-variance decompositions effects differently on the total system mechanism.

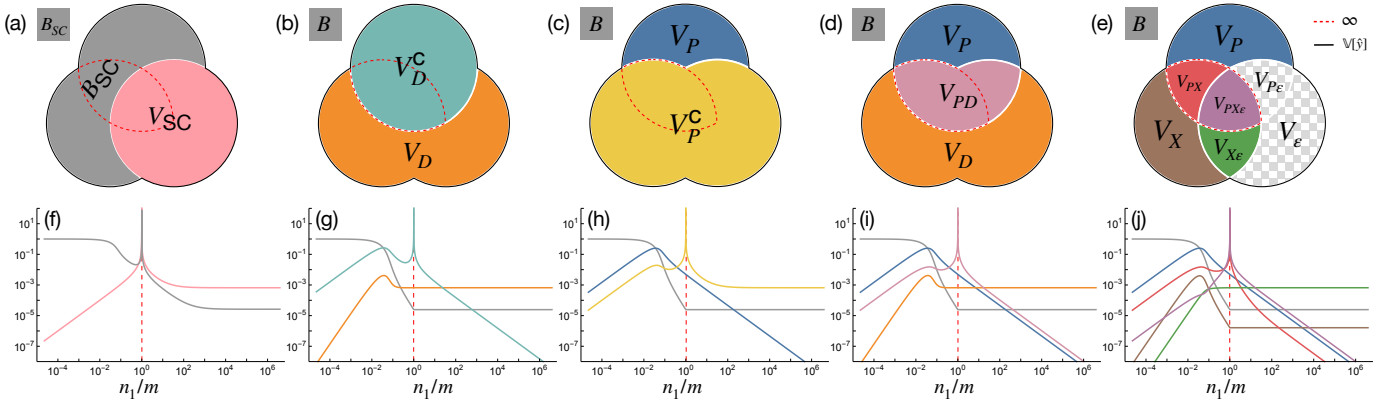


Figure 13. (a-e) The different bias-variance decompositions. (f-j) Corresponding theoretical predictions for $\gamma = 0$, $\phi = 1/16$ and $\sigma = \tanh$ with $\text{SNR} = 100$ as the model capacity varies across the interpolation threshold (dashed red). (a,f) The semi-classical decomposition of Hastie et al. 2019; Mei and Montanari 2019b has a nonmonotonic and divergent bias term, conflicting with standard definitions of the bias. (b,g) The decomposition of Neal et al. 2018 utilizing the law of total variance interprets the diverging term V_D^C as “variance due to optimization”. (c,h) An alternative application of the law of total variance suggests the opposite, i.e. the diverging term V_P^C comes from “variance due to sampling”. (d,i) A bivariate symmetric decomposition of the variance resolves this ambiguity and shows that the diverging term is actually V_{PD} , i.e. “the variance explained by the parameters and data together beyond what they explain individually.” (e,j) A trivariate symmetric decomposition reveals that the divergence comes from two terms, V_{PX} and $V_{PX\epsilon}$ (outlined in dashed red), and shows that label noise exacerbates but does not cause double descent. Since $V_\epsilon = V_{P\epsilon} = 0$, they are not shown in (j). Taken from Adlam and Pennington 2020

The main term of a gradient descent algorithm is the gradient of the loss function $\mathcal{L}(h(x), y)$. For the hypothesis h , the number of instances supplied denoted by k has three main cases: $k = 1$ for stochastic descent, $k \in (1, m)$ for batch gradient descent, and $k = m$ for standard gradient descent, for a given space of m samples. The gradient in such case is calculated using the empirical risk on m size, that is:

$$\mathcal{L}_k(h(x), y) = \hat{R}_{S[k]}(h) = \frac{1}{k} \sum_{i=1}^k \ell(h(x_i), y_i) \quad (44)$$

We use the classical bias-variance tradeoff at first. Since the gradient is a linear operator, we have:

$$\begin{aligned} \nabla_k(\mathcal{L}) &= \nabla_k \left((\mathbb{E} \hat{y}(\mathbf{x}) - \mathbb{E} y(\mathbf{x}))^2 + \mathbb{V}[\hat{y}(\mathbf{x})] + \mathbb{V}[y(\mathbf{x})] \right) \\ &= \nabla_k (\mathbb{E} [\hat{y}(\mathbf{x})] - \mathbb{E} [y(\mathbf{x})])^2 + \nabla_k \mathbb{V}[\hat{y}(\mathbf{x})] + \nabla_k \mathbb{V}[y(\mathbf{x})] \\ &= \nabla_k (\mathbb{E} [\hat{y}(\mathbf{x})] - \mathbb{E} [y(\mathbf{x})])^2 + \nabla_k \mathbb{V}[\hat{y}(\mathbf{x})] \end{aligned} \quad (45)$$

This effectively decompose the gradient into three parts that influence the overall gradient calculation. Because $\nabla_k \mathbb{V}[y(\mathbf{x})]$ calculates the gradient of irreducible error, which is supposed to be constant of the sample space, it is then equal 0. Suppose a sample $\mathbf{x}_k$ of size km , of partition k . Hence, we have gradient descent based on k th run over the entire dataset, permuted. Then, the loss function calculates in the form

$$\nabla_{k,n}(\mathcal{L}) = \nabla_{k,n} \left[\frac{1}{n} \sum_{i \leq n} \hat{y}(x_{kn+i}) - \frac{1}{n} \sum_{i \leq n} y(x_{kn+i}) \right]^2 + \nabla_{k,n} \left[\frac{1}{n} \sum_{i=1}^n y(x_i)^2 - \left(\frac{1}{n} \sum_{i=1}^n y(x_i) \right)^2 \right] \quad (46)$$

The indices is now changed to (k, n) for $m/k = n$ as the i th partitioned, ordered run, thereby $kn + i$ runs for the index. If we assume a simplified structure θ_i of parameters $\gamma(x_i)$ depends on, we can reduce it further.

$$\nabla_{k,n}(\mathcal{L}) = 2 \left(\frac{1}{n} \sum_{i \leq n} \hat{\gamma}(x_{kn+i}) - \frac{1}{n} \sum_{i \leq n} \gamma(x_{kn+i}) \right) \left(\frac{1}{n} \sum_{i \leq n} \nabla_k \hat{\gamma}(x_{kn+i}) - \frac{1}{n} \sum_{i \leq n} \nabla_k \gamma(x_{kn+i}) \right) \quad (47)$$

$$+ \frac{2}{n} \sum_{i=1}^n \gamma(x_i) \nabla_k \gamma(x_i) - \frac{2}{n^2} \left(\sum_{i=1}^n \gamma(x_i) \right) \left(\sum_{i=1}^n \nabla_k \gamma(x_i) \right) \quad (48)$$

$$= 2 \left(\left(\mathbb{E}_n[\hat{Y}] - \mathbb{E}_n[Y] \right) \left(\mathbb{E}_n[\hat{G}_k] - \mathbb{E}_n[G_k] \right) + \mathbb{E}_n[Y G_k] - \mathbb{E}_n[Y] \mathbb{E}_n[G_k] \right) \quad (49)$$

$$= 2 \left(\left(\mathbb{E}_n[\hat{Y}] - \mathbb{E}_n[Y] \right) \left(\mathbb{E}_n[\nabla_k \hat{Y}] - \mathbb{E}_n[\nabla_k Y] \right) + \text{Cov}_n(Y, \nabla_k Y) \right) \quad (50)$$

$$= 2 \left(\mathbb{E}_n[\hat{Y}] - \mathbb{E}_n[Y] \right) \left(\mathbb{E}_n[\nabla_k \hat{Y}] - \mathbb{E}_n[\nabla_k Y] \right) + 2 \text{Cov}_n(Y, \nabla_k Y) \quad (51)$$

for

$$\mathbb{E}_n[Z] = \frac{1}{n} \sum_{i=1}^n Z_i, \quad \text{Cov}_n(U, V) = \mathbb{E}_n[UV] - \mathbb{E}_n[U] \mathbb{E}_n[V]. \quad (52)$$

Here, we can see two terms multiplication, for the first one being the correlation between the expected error $\mathbb{E}_n[\hat{Y}] - \mathbb{E}_n[Y]$ and the gradient $\mathbb{E}_n[\nabla_k \hat{Y}] - \mathbb{E}_n[\nabla_k Y]$. This measures the correlation between the expected error, and the gradient measure. The second term is exactly the empirical covariance between the target Y and the true gradient component. Since we are taking a fixed concept case, we can simply disregard the truth gradient hence $\mathbb{E}_n[\nabla_k Y]$, for which the term is reduced to $2 \left(\mathbb{E}_n[\hat{Y}] - \mathbb{E}_n[Y] \right) \left(\mathbb{E}_n[\nabla_k \hat{Y}] \right)$. Under this same assumption applied to the covariance term, we obtain

$$\nabla_{k,n}(\mathcal{L}) = \frac{2}{n} \sum_{i=1}^n (\hat{Y}_i - Y_i) \partial_k \hat{Y}_i = 2 \left(\left(\mathbb{E}_n[\hat{Y}] - \mathbb{E}_n[Y] \right) \mathbb{E}_n[\nabla_k \hat{Y}] \right) \quad (53)$$

Looking into some particular algebraic representation, we can write this simply as

$$\nabla_{k,n} \mathcal{L} = 2 \mathbb{E}_n \left[(\hat{Y} - Y) \nabla_k \hat{Y} \right] \quad (54)$$

$$= 2 \left(\underbrace{\left(\mathbb{E}_n[\hat{Y}] - \mathbb{E}_n[Y] \right) \mathbb{E}_n[\nabla_k \hat{Y}]}_{[\mathbb{B}(\hat{Y}, Y)]^{1/2}} + \underbrace{\mathbb{E}_n[\hat{Y} \nabla_k \hat{Y}] - \mathbb{E}_n[\hat{Y}] \mathbb{E}_n[\nabla_k \hat{Y}]}_{\text{Cov}_n(\hat{Y}, \nabla_k \hat{Y})} \right) \quad (55)$$

$$- \underbrace{\mathbb{E}_n[Y \nabla_k \hat{Y}] - \mathbb{E}_n[Y] \mathbb{E}_n[\nabla_k \hat{Y}]}_{\text{Cov}_n(Y, \nabla_k \hat{Y})} \quad (56)$$

The first term can be attributed to self-correct of bias-error gradient correlation, while the two covariance term suggests correlation between model randomness and its gradient, and the last one deal with the truth-gradient correlation. What those gradients mean will have to be tested in some preliminary models. Generally, we should not expect the result to be quite effective. Similar decomposition can be conducted for the other decomposition, as suggested, however, the gradient effect is particularly interesting, and will have to be resolved.

Appendix C - Details of supervised learning structure

In a typical machine learning setting, we obtain assume the following figure 14. This includes the ground, input space (the setting of which the observations are observed), the concept $c \in \mathcal{C}$, the hypothesis $c \in \mathcal{H}$, and a supervisor ∇ that 'correct' the behaviours of the hypothesis to match with c . This is called a supervised learning setting, in which the learning part is governed by the supervisor and the observations from the concept.

Phase 1.

Begin with the construction and initialization of the feature space $\mathcal{X}^\infty$. By the assumption of a probabilistic process, $\mathcal{X}_m$ or simply $\mathcal{X}$ representing the dataset is assumed to be sampled, or appeared from the distribution $\mathcal{D}(p, \cdot)$ for p an arbitrary probabilistic system.

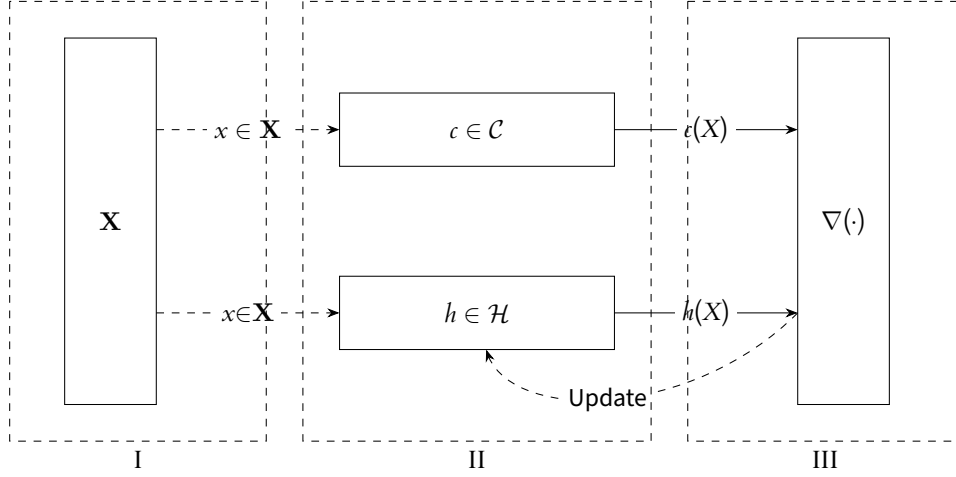


Figure 14. An illustration of the (supervised) statistical process. Phase III contains two parts: First is the evaluation $\nabla(h, c)$ according to the data $\mathcal{D}$, and second is the Update process to re-align c to the actual target.

The *ground space*, or in literature generally called **feature space**, is created. This results in the first component of the tuple D , being $\mathcal{X} \subseteq \mathbb{R}^{m \times k}$, where $|\mathcal{X}| = m$, but $\text{size}(x) = k$, for $x \in \mathcal{X}$. We assume there exists the random variable x of a given distribution $\mathcal{D}$, such that the set $\mathcal{X} \sim \mathcal{D}$ of all observation is given by an underlying distribution. A well-known assumption in this case is that $x \in \mathcal{X}$ has no dependent components. That is, for example, there does not exist any function $\psi_x : \{x_i\} \rightarrow \{x_j\}$, for $i \neq j$. (Thought, this only helps in Bayesian priori analysis). Furthermore, by specifying that $\mathcal{X}$ belongs to a specific distribution, we argue of the assumption that $x \in \mathcal{X}$ is *independently and identically distributed* (i.i.d.), that is, for a given random sampling interpretation $S \subset \mathcal{X} \sim \mathcal{D}$, all data is sampled with the events space governed by the distribution $\mathcal{D}$ for every instance, and the existence of x_i to x_j for $i \neq j$ is none.

The **role** of the distribution configuration is important. If $\mathcal{D}$ is uniform, that is, $D \sim \mathcal{U}(a, b)$, then we can disregard the aspect of $\mathcal{X}$ with random variables. For $[a, b]$ to be $(-\infty, +\infty)$, the problem changed to a *regression problem*. If $\mathcal{D}$ is some other distribution function, then the problem of *representative data* and *population size* becomes apparent, which requires statistical analysis to be taken into consideration.

Phase 2.

For the constructed feature space $\mathcal{X}$, let $EX(c, \mathcal{D})$ be a procedure (or *oracle*) acting on $\mathcal{C}$ and $\mathcal{X}$ to output $\langle x, c(x) \rangle$. The hypothesis then contains a similar procedure $EX_{\mathcal{H}}(h, \mathcal{D})$, such that to output $\langle x, h(x) \rangle$. We call this the *inference phase*.

The feature space $\mathcal{X}$ is provided to h . We assume c is processed of the same process, resulted in $\mathcal{Y}_c$. Then, $h(\mathcal{X})$ outputs the set of hypothesis' target $\mathcal{Y}_h$. Thereby, we are approximating the *process* c itself. We can approach this in two main ways^l: either *deterministic* or *probabilistic*^m. We define such as followed. For the constructed feature space $\mathcal{X}$, let $EX(c, \mathcal{D})$ be a procedure (or *oracle*) acting on $\mathcal{C}$ and $\mathcal{X}$ to output $\langle x, c(x) \rangle$. The hypothesis then contains a similar procedure $EX_{\mathcal{H}}(h, \mathcal{D})$, such that to output $\langle x, h(x) \rangle$. We call this the *inference phase*.

Definition Appendix 0.1 (Deterministic – discriminative modelling). A **deterministic model** h_d assumes its internal space as followed. For any $h \in \mathcal{H}_d$ of deterministic model, h is characterized by the mapping $h_d : \mathcal{X} \rightarrow \mathcal{Y}_h$, such that $h \in C^n$ of n -differential space.

Similarly, if the interpretation of the model is *probabilistic*, we have the following definition of probabilistic-based model.

Definition Appendix 0.2 (Probabilistic – generative modelling). A **probabilistic model** h_p assumes its internal space as followed. For any $h \in \mathcal{H}_p$ of all probabilistic hypothesis, h is characterized by the probability distribution $\Gamma(p_X(X))$, where Γ is the discrete decision process outside the probabilistic interpretation.ⁿ

^l. Note that, in some aspect, this is similar to the usage of mathematical simulation of certain dynamical system, or *any* system that has certain patterns or relations observable.

^m. The model itself can be entirely deterministic, while the learning process is not. in fact, one of the very first assumption and axiomatic view of the learning problem is that learning occurs in a *probabilistic setting*.

ⁿ. The discrete decision process Γ is very simple to argue. For example, given a Naive Bayes model with $p(C_k | D[i])$ of the dataset, the probability distribution is regarded as the argument

$$Y^* = \arg \max_{c_k \in C} P_{\Theta}(c_k | x) = \arg \max_{c_k \in C} P_{\Theta}(x | c_k) \cdot P(c_k)$$

Notice that the process and the learning setting is probabilistic, however, the interpretation of the hypothesis h can be either deterministic or probabilistic, or anything else. This puts the flexibility into the setting of the model, and permits us to consider various representation of the model structure. Furthermore, the hypothesis for the learning process is evaluated relative to the same probabilistic setting, in which the training takes place, and we allow hypotheses that are only approximation of the target concept Sugiyama 2015; Kearns and Vazirani 1994.

The similarity between both of the two general types is that the model is characterized by their **parameters**, both **model-specific parameters** and **hyperparameters**.

Phase 3.

There exists an evaluator (or *supervisor*) $\nabla(h, c)$ that evaluates specific *loss framework* $\ell\{h(x), c(x)\}$ based on available information, and a *update modifier* $U(\ell, \mathcal{A})$ of certain objective to algorithm $\mathcal{A}$.

In this step of the procedural setting, the *supervisor* includes a loss function, ℓ , which takes discrete arguments, and ‘binarily’ evaluate the discrete result between h and c . This loss function is supposed to have a global minimum, or at least a certain region of minimal approximation. We have the following definition.

Definition Appendix 0.3 (Loss function). *A loss function is a non-negative function $\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow [0, +\infty)$.*

In general, the following is supposed of the loss function:

Conjecture Appendix 0.1 (Loss function convexity). *For any supervisor ∇ of a learner framework $\mathcal{L}(\mathcal{H}, \mathcal{A})$, the loss function ℓ is assumed to be a p -converging function, or as a convex function. That is, for $\ell : \mathbb{R}^n \rightarrow \mathbb{R}$, then ℓ is convex on its domain, or is locally convex on $[a, b] \subset \mathbb{R}^n$, if, for $x, y \in \mathbb{R}^n$ of such interval, and for $\lambda \in [0, 1]$, it satisfies*

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y) \quad (57)$$

Under a single structure, that is, for example, a class of mapping $\mathbb{R}^n \rightarrow \mathbb{R}$, there can exist many loss functions to be considered. Take for example the setting of regression analysis. Then, ℓ can be either the absolute error, $|h(x) - c(x)|$, or the L^2 norm, $\|h(x) - c(x)\|_2$. Different loss function provides different path and landscape, though they still share somewhat similar interpolation patterns, or either some might be subjected to, under the consideration of a loss landscape, local minima than others. Furthermore, the form and representation of loss functions is not discrete – but rather based of its interpretation and the supposition of the underlying notion required for the loss function to operate in. For example, if the setting is *generative*, the loss function would take the form of a *divergence measure* between two probability distribution. Notice however, that the action potential provided by the model in their operation dissappears, or rather, is not considered, when using the loss function.

Using such loss function, for the case of no noises or interference uncertainty, if the goal is to minimize the empirical risk on such dataset, then we call this the method of *empirical risk minimization*, or ERM. This is born out of the consideration that in practice, the oracle $EX(c, \mathcal{D})$, and the actual generalization error $R(h)$ is not obtainable.

Appendix D - Bias-variance decomposition derivation in Geman, Bienenstock, and Doursat 1992

Proof. We present the original derivation of bias-variance tradeoff and its analysis from Geman, Bienenstock, and Doursat 1992. Geman’s paper is, by his own word, concerned of *parametric models*, i.e. hypothesis class without strong assumption of parameters defined, though the definition of bias-variance tradeoff afterward and its justification is made in the sense of parametric model. We partially clarify^o this with the following definition as customary.

Definition Appendix 0.4 (Parameterization). *A **parametric model** is one that can be parameterized by a finite number of parameters. In general, for a hypothesis class $\mathcal{H}$ of all parametric hypothesis is then expressed as:*

$$\mathcal{H} = \{f(x; \theta) : \theta \in \Theta \subset \mathbb{R}^d\} \quad (58)$$

where Θ is called the parameter space. A **nonparametric model** is one which cannot be parameterized by a fix number of parameters.

Suppose of a training dataset $\mathcal{D}$ of N 2-tuples $(\mathbf{x}_i, y_i)$, that is:

$$\mathcal{D} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_N, y_N)\} \quad |\mathcal{D}| = N, \mathbf{x} \in \mathbb{R}^d, y \in \mathbb{R} \quad (59)$$

^o. Generally, there is no definite definition on what would be considered non-parametric. Though, an example might be drawn from the (vanilla) neural network and Gaussian Process Regression (GPR)

The regression problem is to construct a function $f(\mathbf{x}) \rightarrow y$ based on $\mathcal{D}$, which is denoted $f(\mathbf{x}; \mathcal{D})$ to show this dependency. Then,

$$\begin{aligned}
 \mathbb{E} \left[(y - f(\mathbf{x}; \mathcal{D}))^2 \mid \mathbf{x}, \mathcal{D} \right] &= \mathbb{E} \left[((y - \mathbb{E}[y \mid \mathbf{x}] + (\mathbb{E}[y \mid \mathbf{x}] - f(\mathbf{x}; \mathcal{D})))^2 \mid \mathbf{x}, \mathcal{D} \right] \\
 &= \mathbb{E} \left[(y - \mathbb{E}[y \mid \mathbf{x}])^2 \mid \mathbf{x}, \mathcal{D} \right] + \left(\mathbb{E}[y \mid \mathbf{x}] - f(\mathbf{x}; \mathcal{D}) \right)^2 \\
 &\quad + 2\mathbb{E} \left[(y - \mathbb{E}[y \mid \mathbf{x}]) \mid \mathbf{x}, \mathcal{D} \right] \cdot (\mathbb{E}[y \mid \mathbf{x}] - f(\mathbf{x}; \mathcal{D}))^2 \\
 &= \mathbb{E} \left[(y - \mathbb{E}[y \mid \mathbf{x}])^2 \mid \mathbf{x}, \mathcal{D} \right] + (\mathbb{E}[y \mid \mathbf{x}] - f(\mathbf{x}; \mathcal{D}))^2 \\
 &\geq \mathbb{E} \left[(y - \mathbb{E}[y \mid \mathbf{x}])^2 \mid \mathbf{x}, \mathcal{D} \right]
 \end{aligned} \tag{60}$$

Hence, we decompose it to:

$$\mathbb{E} \left[((y - f(\mathbf{x}; \mathcal{D})))^2 \mid \mathbf{x}, \mathcal{D} \right] = \mathbb{E} \left[(y - \mathbb{E}[y \mid \mathbf{x}])^2 \mid \mathbf{x}, \mathcal{D} \right] + (f(\mathbf{x}; \mathcal{D}) - \mathbb{E}[y \mid \mathbf{x}])^2 \tag{61}$$

Where $\mathbb{E} \left[(y - \mathbb{E}[y \mid \mathbf{x}])^2 \mid \mathbf{x}, \mathcal{D} \right]$ is regarded to be a relative constant of variance on y given $\mathbf{x}$. Taking the expectation on $\mathcal{D}$, we gain:

$$\begin{aligned}
 \mathbb{E}_{\mathcal{D}} \left\{ \mathbb{E} \left[((y - f(\mathbf{x}; \mathcal{D})))^2 \mid \mathbf{x}, \mathcal{D} \right] \right\} &= \mathbb{E}_{\mathcal{D}} \left\{ \mathbb{E} \left[(y - \mathbb{E}[y \mid \mathbf{x}])^2 \mid \mathbf{x}, \mathcal{D} \right] + (f(\mathbf{x}; \mathcal{D}) - \mathbb{E}[y \mid \mathbf{x}])^2 \right\} \\
 &= \mathbb{E}_{\mathcal{D}} \left\{ \mathbb{E} \left[(y - \mathbb{E}[y \mid \mathbf{x}])^2 \mid \mathbf{x}, \mathcal{D} \right] \right\} + \\
 &\quad \mathbb{E}_{\mathcal{D}} \left\{ (f(\mathbf{x}; \mathcal{D}) - \mathbb{E}[y \mid \mathbf{x}])^2 \right\}
 \end{aligned} \tag{62}$$

The second term is of importance, and is further decomposed to:

$$\begin{aligned}
 &\mathbb{E}_{\mathcal{D}} \left\{ (f(\mathbf{x}; \mathcal{D}) - \mathbb{E}[y \mid \mathbf{x}])^2 \right\} \\
 &= \mathbb{E}_{\mathcal{D}} \left\{ (f(\mathbf{x}; \mathcal{D}) - \mathbb{E}_{\mathcal{D}}[f(\mathbf{x}; \mathcal{D})] + \mathbb{E}_{\mathcal{D}}[f(\mathbf{x}; \mathcal{D})] - \mathbb{E}[y \mid \mathbf{x}])^2 \right\} \\
 &= \mathbb{E}_{\mathcal{D}} \left[(f(\mathbf{x}; \mathcal{D}) - \mathbb{E}_{\mathcal{D}}[f(\mathbf{x}; \mathcal{D})])^2 \right] + \mathbb{E}_{\mathcal{D}} \left[(\mathbb{E}_{\mathcal{D}}[f(\mathbf{x}; \mathcal{D})] - \mathbb{E}[y \mid \mathbf{x}])^2 \right] + \\
 &\quad 2\mathbb{E}_{\mathcal{D}} \left[(f(\mathbf{x}; \mathcal{D}) - \mathbb{E}_{\mathcal{D}}[f(\mathbf{x}; \mathcal{D})]) \cdot (\mathbb{E}_{\mathcal{D}}[f(\mathbf{x}; \mathcal{D})] - \mathbb{E}[y \mid \mathbf{x}]) \right] \\
 &= \underbrace{\mathbb{E}_{\mathcal{D}}[f(\mathbf{x}; \mathcal{D})] - \mathbb{E}[y \mid \mathbf{x}]}_{\text{bias term}} + \underbrace{\mathbb{E}_{\mathcal{D}} \left\{ (f(\mathbf{x}; \mathcal{D}) - \mathbb{E}_{\mathcal{D}}[f(\mathbf{x}; \mathcal{D})])^2 \right\}}_{\text{variance term}}
 \end{aligned} \tag{63}$$

which yields the desired bias-variance tradeoff. $\square$